Typed Responsibility Licensing for Scientific Revision: Transition Envelopes and Exact Mechanism Evidence

Working framework draft

August 2026

Abstract

A successful diagnosis or repair does not determine which scientific layer the evidence authorizes changing. We formalize typed responsibility-to-authority licensing through epistemic transition envelopes, exact interface factorization, and legal-action deficiency. In prospectively frozen primary and disjoint generator runs of 2,882 mechanical worlds each, the licensed-transition policy attained hidden-shift success 1.0000 with no unnecessary high-level changes on 2,402 controls per run. A later faithful-comparator study shows that a parent granted the same ordered search also reaches 1.0000 with zero forbidden mutations, so no comparative mechanism-necessity margin is claimed; the surviving comparative residual is intervention-cost economy at equal success and safety. On 400 authored exact contracts, it made 400 correct decisions versus 275 for a donor-complete interface lacking the licensing coupling; an information-equivalent product tied at 400/400. These results establish a finite mechanism boundary, not naturalistic or inherent-expressivity superiority. The historical 48-case precursor remains negative. In the naturalistic programme, 117,649 source-to-target maps yield 116,929 rejections, 720 unresolved maps, and zero certifications. V13 makes external authority execution content-addressed but has received 0/7 signed outputs and closed 0/4 authority acts.

Keywords: epistemic transitions; scientific reformulation; information sufficiency; belief revision; autonomous science; research agents.

1 Introduction

1.1 The failure mode

A research process can make progress while its own framing becomes the dominant source of error. More search does not repair a missing representation. A larger tree does not by itself reveal that the wrong parent discipline has been treated as irrelevant. More evidence does not justify rewriting a question when the actual defect is merely an execution failure. The problem studied here is therefore not generic iteration, autonomous science, or long-horizon reasoning. It is the narrower governance problem of preserving an inspectable epistemic state while deciding when evidence is strong enough to authorize a change in the scientific formulation itself.

For a scoped problem, the architecture studied here represents the evolving research state as

$$X_t = (K_t, W_t, M_t),$$

where K_t is the provenance-preserving state of knowledge about the target object, W_t is the current model of which domains, representations, mechanisms, and search routes may be relevant, and M_t is the governed research method. The three coordinates are separated because they require different evidence and different authority to change. This engineering separation is in the spirit of Parnas's information-hiding principle: change-prone design decisions should be isolated behind explicit interfaces rather than entangled implicitly [35]. The revision policy for K_t likewise borrows the minimal-change distinction between expansion, contraction, and revision from the AGM belief-revision tradition [2]; no claim is made here that the operational update rules satisfy the full AGM postulates, which would have to be established separately.

A new source may update K_t without changing W_t. A previously unrecognized parent discipline or representation may require a change to W_t while leaving the method intact. A persistent, well-attributed failure can become evidence for a challenger to M_t, but method revision is governed separately and is not an operation this paper authorizes.

1.2 Protected scientific contribution

Paper I isolates one dominant scientific contribution with four coupled requirements. First, $K/W/M$ are explicit co-evolving state rather than hidden conversational context. Second, discoveries and failures target typed formulation coordinates, so diagnosed responsibility constrains which high-level mutation is admissible. Third, high-level change is gated by lower-level exclusion, same-task counterfactual necessity, and protected-invariant preservation rather than generic "reflect and retry." Fourth, an accepted mutation stales exactly the dependent closure identified by its bound impact rather than allowing prior success certificates to survive silently.

1.3 Contributions

Four properties of the mechanism are put forward and tested.

The key advance is the coupling. Recursive audit, active diagnosis, counterfactual repair, dependency rollback, certificate enforcement, evidence-gated revision, and generic mutation authorization are retained as strong donor mechanisms. ORION composes them into a protected scientific-escalation contract in which responsibility determines the candidate mutation, counterfactual necessity tests whether high-level change is actually needed, protected preservation prevents collateral regression, and dependency impact determines exactly what must reopen.

The powered hidden-shift primary/replication study tests whether that contract can make reformulation both successful and selective. The P1-X successor then generalizes the same control law to heterogeneous revision-responsibility contracts after a donor-complete comparison product is granted the underlying mechanisms. Together, these studies test the same architectural object at two levels rather than distributing one result across unrelated claims.

The resulting scientific claim is direct: coupling responsibility, counterfactual necessity, protected preservation, and dependency impact defines an explicit decision contract for high-level scientific reformulation, and the prospectively frozen studies show full protected recovery while maintaining exact restraint on the registered control families. Comparative gains over the registered strong parents are withdrawn; see Section 6.

1.4 Five-paper boundary

Paper I owns the reconstruction problem. Open-world discovery and stopping are evaluated in Paper II; cross-domain absorption and the global knowledge portrait in Paper III; authority promotion and protected verification in Paper IV; and method evolution in Paper V. Paper I uses those components as explicit interfaces so that the reformulation result remains attributable to one well-defined scientific object rather than to an umbrella architecture. What is defended empirically is narrower than any of the four taken alone: it is the composition of protected necessity conditions under which a high-level formulation or search-universe change is admissible at all. Section 10 places each ingredient with its nearest neighbour in the literature, and Section 12 states what the composition is and is not evidence for.

2 Recursive reconstruction protocol

The minimum research cycle is

$$\begin{aligned}\text{FRAME} \to \text{SEARCH} \to \text{ABSORB} \to \text{RECONSTRUCT} \to \text{DETECT}\\ \to \text{DIAGNOSE} \to \text{REFRAME} \to \text{REOPEN} \to \text{RECURSE.}\end{aligned}$$

Frame. The current question, scope, decomposition, observables, and success conditions are made explicit enough that later changes can be attributed to named coordinates rather than to an opaque prompt.

Search and absorb. Search produces candidate material but has no scientific-authority write capability. Absorption maps material into typed contributions and provenance-bearing state. Stronger discovery and verification contracts are outside the scope of this paper; what reconstruction needs is only the non-escalation invariant that a proposal or retrieval event cannot make itself authoritative.

Reconstruct. The current state is reconstructed from retained evidence, mappings, assumptions, interfaces, unresolved coordinates, and prior decisions. Reconstruction is not simply summarization. It must preserve enough lineage to answer why a coordinate has its current value and what prior closure depends on it.

Detect and diagnose. Residuals are typed before corrective action. A missing source, an execution defect, a representation mismatch, a search-universe omission, and a method defect are not interchangeable. Diagnosis must therefore identify the supported responsibility level and preserve competing explanations when the evidence is insufficient to distinguish them.

Reframe. Only the coordinates supported by the diagnosis may change. In particular, evidence or execution failure does not by itself license a formulation rewrite. Method-state changes are governed separately and are not authorized here.

Reopen. A material change invalidates dependent closure. Nothing is reset indiscriminately; what reopens is the set of conclusions, search obligations, or mechanic contracts whose validity depended on the changed coordinate.

Recurse. Research resumes from the new state. The next cycle must be able to compare the reconstructed state with the previous one, preserving negative history rather than treating a reframe as a fresh conversation with no memory.

2.1 Authority separation

A proposer, language model, query generator, or search engine can be computationally central while remaining epistemically non-authoritative. This distinction is load-bearing for reconstruction: if a component can both propose a new formulation and silently promote it to scientific fact, the state transition cannot be meaningfully audited. Proposal, evidence acquisition, diagnosis, revision, and verification are therefore distinct roles even when one provider happens to implement several of them.

3 Epistemic transition envelopes

The general object studied in this paper is not a particular diagnosis, repair, rollback, or belief-revision component. It is an *epistemic transition envelope*: the admissible frontier of scientific changes after those components have supplied their outputs. The envelope separates three questions that are often collapsed: which transitions could restore performance, which transitions the evidence authorizes, and which authorized transition is minimal in the scientific change order. The $K/W/M$ architecture used in the experiments is one realization of this object, not part of its mathematical definition.

Mathematical conventions. Unless a probability law is explicitly introduced, the equalities and factorizations in this section are pointwise, set-theoretic statements on the declared context class. They do not by themselves assert that a factor map is measurable or computable; a measurable version additionally requires the corresponding quotient/factor map to be measurable. Once a law is introduced, conditional distributions and their conclusions are defined only almost surely, so a null interface fibre receives no pointwise certification. Every decision alphabet used below is nonempty. For the Bayes result we use standard Borel spaces and a finite discrete decision alphabet, so regular conditional laws and a measurable fixed-order maximizer exist.

3.1 Contexts, transition orders, and admissible fronts

Let $\mathcal{E}$ be an arbitrary nonempty set of evidence contexts. A context e has a candidate transition space $\mathcal{R}_e$ and a strict partial order $\prec_e$. The interpretation of $r' \prec_e r$ is that r' makes strictly less scientific change than r in context e. The order need not be a chain: revisions of a measurement model and a problem boundary, for example, can be incomparable. We require only that the order restricted to any admissible set encountered below be well founded. This includes every finite revision space but also permits infinite spaces without infinite descending chains.

Let $C_{1,e}, \ldots, C_{k,e}$ be hard admissibility predicates. Their conjunction, admissible set, and minimal front are

$$C_e(r) = \bigwedge_{i=1}^{k} C_{i,e}(r), \qquad \mathcal{A}(e) = \{r \in \mathcal{R}_e : C_e(r)\}, \qquad \mathcal{F}(e) = \min_{\prec_e} \mathcal{A}(e).$$

For the scientific-escalation instance evaluated here, $k = 5$ and

$$\begin{aligned}
J_e(r) &: \text{ the evidence licenses revision at the layer changed by } r,\\
T_e(r) &: \ r \text{ restores the same registered task under intervention,}\\
P_e(r) &: \ r \text{ preserves every registered sibling or invariant,}\\
D_e(r) &: \text{ observed dependency impact equals the declared reopen scope,}\\
B_e(r) &: \ r \text{ respects the registered action and resource budget.}
\end{aligned}$$

Other scientific domains may replace or add hard predicates without changing the results below.

Let $Q(e)$ say that the registered task has a material residual. The transition decision Γ is

$$\Gamma(e) = \begin{cases} \mathsf{NO\_CHANGE}, & \neg Q(e), \\ \mathsf{UNRESOLVED}, & Q(e) \text{ and } \mathcal{A}(e) = \varnothing, \\ r, & Q(e) \text{ and } \mathcal{F}(e) = \{r\}, \\ \mathsf{MEASURE}(\mathcal{F}(e)), & Q(e) \text{ and } |\mathcal{F}(e)| > 1. \end{cases}$$

The last output carries the unresolved front rather than hiding an arbitrary tie-break. A later discriminating observation may refine the context and order. Thus abstention and further measurement are identified scientific actions, not missing values silently converted to failure or success.

Proposition 1 (sound, non-compensatory minimality). If $\Gamma(e) = r \in \mathcal{R}_e$, then every $C_{i,e}(r)$ holds and there is no $r' \in \mathcal{A}(e)$ with $r' \prec_e r$. Moreover, optimizing any utility, accuracy, or cost score over $\mathcal{F}(e)$ cannot select a transition that fails a hard predicate, regardless of its score.

Proof. The first two statements are the definition of membership in $\mathcal{F}(e)$. The last follows from $\mathcal{F}(e) \subseteq \mathcal{A}(e)$: candidates outside the conjunction are not in the optimization domain and cannot compensate for a failed gate. □

This is the minimal-escalation safety property tested by the mechanical studies. The broader envelope formalism, however, is not tied to five gates, three epistemic layers, or a finite generator.

3.2 Exact substitution is a factorization property

Suppose an arbitrary product of donor mechanisms exposes an interface $\phi : \mathcal{E} \to \mathcal{Z}$. The interface can combine outputs from diagnosis, repair, belief revision, dependency maintenance, model discovery, planning, or human review. A downstream policy sees $\phi(e)$ but not e. Write $\mathcal{Y}$ for the output space of Γ.

Theorem 1 (exact donor-substitution theorem). The following are equivalent:

1. there exists a policy $g : \phi(\mathcal{E}) \to \mathcal{Y}$ such that $\Gamma = g \circ \phi$;
2. Γ is constant on every fibre of ϕ, that is,
$$\phi(e) = \phi(e') \quad \Longrightarrow \quad \Gamma(e) = \Gamma(e');$$
3. the partition of contexts induced by ϕ refines the partition induced by the required transition decision.

Proof. If $\Gamma = g \circ \phi$, equal interface values give equal decisions, so (1) implies (2). If (2) holds, define $g(z) = \Gamma(e)$ for any e with $\phi(e) = z$. Fibre constancy makes this definition independent of the chosen representative, proving (1). Statement (3) is the partition form of (2). □

The theorem gives a necessary and sufficient condition, not merely the sufficient condition that a donor product reconstruct the full tuple $(\mathcal{R}_e, \prec_e, C_{1,e}, \ldots, C_{k,e}, Q(e))$. Full reconstruction guarantees equivalence, but a smaller statistic is enough whenever it separates all contexts requiring different decisions. Conversely, component count, mechanism sophistication, and successful repair rates cannot compensate for a mixed decision fibre. This is the precise sense in which an envelope can absorb the strongest available donor mechanisms while still identifying a missing interface relation.

Corollary 1 (information-equivalent products must tie). Any donor product that reconstructs the transition tuple above and applies the same decision rule is extensionally identical to Γ on every context. Reconstruction may use a relabelled representation only when it comes with a known decision-preserving isomorphism: the alignment must preserve candidate correspondence, preserve and reflect the strict transition order and every predicate truth value, preserve $Q(e)$, and map each reconstructed front/output back to the original Γ decision. Equal component counts, or an unaligned isomorphism of otherwise anonymous encodings, is not information equivalence in this sense.

The ideal-product tie in Section 7 is therefore a required negative boundary. It is evidence against an inherent centralization or expressivity advantage, not an inconvenient null result.

Terminal-transport certification boundary. Exact set-theoretic factorization does not manufacture target semantics. An outcome-blind audit compared six source-native postpublication terminals with seven inherited process-decision identifiers. Among all $7^6 = 117{,}649$ total maps, none is fully certified by the inherited semantics; this does not imply that no semantics-preserving map exists. Because the six actionable target decisions lack exhaustive coordinate, licensed-operation and forbidden-operation denotations, and because terminal decisions and diagnostic actions have no inherited one-to-one bridge, 720 injective maps remain not disproved but uncertified. The only currently certified status-preserving partial mapping sends UNRESOLVED to UNRESOLVED; every actionable image remains unbound. Full-artifact withdrawal is an impossibility witness only conditional on later owner-ratification of the proposed closed-world target denotations. This is an additional typed-transport gate, not a corollary of Theorem 1.

3.3 Legal-action deficiency of a lossy scientific interface

Exact equivalence is a worst-case property. A distributional version must also respect unequal harms and context-dependent legal actions. Let $E \sim \mu$ on a standard Borel context space, let $Z = \phi(E)$ take values in a standard Borel interface space, and let $\mathcal{A}_0$ be a finite nonempty action alphabet. Each context has a nonempty legal set $\mathcal{A}(E) \subseteq \mathcal{A}_0$ and a nonnegative measurable loss $L(E, a)$. Define the actions that are legal almost surely on an observed fibre by

$$\mathcal{A}_\phi(z) = \{a \in \mathcal{A}_0 : \Pr(a \in \mathcal{A}(E) \mid Z = z) = 1\}.$$

When this set is nonempty almost surely, define

$$R^*_{\text{full}} = \mathbb{E}\left[\min_{a\in\mathcal{A}(E)} L(E, a)\right],$$
$$R^*_\phi = \mathbb{E}_Z\left[\min_{a\in\mathcal{A}_\phi(Z)} \mathbb{E}\{L(E, a) \mid Z\}\right].$$

Theorem 2 (legal-action interface deficiency). Assume $\mathcal{A}_\phi(Z)$ is nonempty almost surely and the displayed expectations are finite. Every deterministic or randomized policy using only Z and legal almost surely has risk at least R^*_ϕ. A deterministic fibrewise minimizer attains this value, and

$$\Delta_\phi = R^*_\phi - R^*_{\text{full}} \geq 0.$$

Equality holds exactly when almost every observed fibre admits a common legal action that is full-context Bayes-optimal almost surely on that fibre.

Proof. Condition on $Z = z$. Any randomized legal policy is supported on $\mathcal{A}_\phi(z)$, so its conditional risk is a convex combination of the conditional risks of those actions and cannot be smaller than their minimum. Because $\mathcal{A}_0$ is finite, a fixed tie order gives a measurable minimizer and attains R^*_ϕ. For every common legal action a, $L(E, a) - \min_{b \in \mathcal{A}(E)} L(E, b) \geq 0$ almost surely. Conditional expectation, minimization over common legal actions, and then expectation over Z prove $\Delta_\phi \geq 0$. Equality holds exactly when a minimizing common legal action has zero regret almost surely on almost every fibre. Emptiness of $\mathcal{A}_\phi(z)$ is precisely the absence of an action an interface-only policy can select while remaining legal almost surely conditional on that fibre. □

Infeasible-fibre boundary. If $\Pr\{\mathcal{A}_\phi(Z) = \varnothing\} > 0$, the displayed R^*_ϕ is not defined and no Z-measurable policy with actions in $\mathcal{A}_0$ can be legal almost surely. The fail-closed conclusion is then the meta-level status UNRESOLVED/CANNOT CHECK, not an action with an imputed loss. If deferral is intended to be feasible, it must instead be included explicitly in $\mathcal{A}_0$ and made legal in the relevant contexts.

If every action is legal and $L(E, a) = \mathbf{1}\{a \neq \Gamma(E)\}$ for a finite decision alphabet, the theorem reduces to

$$R^*_\phi = 1 - \mathbb{E}_Z \left[\max_a \Pr\{\Gamma(E) = a \mid Z\} \right],$$

the earlier decision-fibre impurity. Thus unequal transition harms, resource-sensitive legality, and unresolved/deferral when declared as an explicit legal action are part of the same interface-deficiency object rather than post-hoc exceptions. This decision-theoretic reading is donor territory related to Blackwell's comparison of experiments [5]. The contribution claimed here is its specialization to typed, hard-gated scientific transitions and the empirical isolation of omitted couplings, not the general comparison of information structures.

Corollary 2 (component-essentiality criterion). Let the complete interface be $S = (S_1, \ldots, S_m)$ and let S_{-j} omit one component. Exact decisions are possible from S_{-j} if and only if no two contexts agree on S_{-j} while requiring different Γ decisions. If a fibre is supported on exactly two decision classes with conditional probability $1/2$ each, every policy omitting S_j has conditional error at least $1/2$ on that fibre. More generally its optimal conditional error is $1 - \max_y \Pr\{\Gamma(E) = y \mid S_{-j}\}$ on the fibre.

This turns every negative ablation into a new scientific question: characterize the mixed fibres it creates, identify which missing relation splits them, and test whether those fibres occur under a naturalistic acquisition distribution. It does not turn a failed outcome into a positive result by relabeling it.

Corollary 3 (conditional licence-coupling impossibility). Consider an interface that exposes candidate revisions, recovery results, invariant checks, dependency impacts, and budgets, but omits J_e, the relation coupling evidence to the layer authorized to change. If the context class contains two situations with the same reduced interface and different Γ decisions because their licence relations differ, no policy using only that interface can agree with Γ on both. Under an equal mixture of such a pair, every policy has error at least $1/2$.

Proof. Take $r_{\mathrm{L}} \prec r_{\mathrm{H}}$. In a context class containing the following pair, hold the candidate, recovery, preservation, impact, and budget outputs identical. In e_0, the evidence licenses only r_{L}. In e_1, a registered exclusion witness licenses only r_{H}. Hence $\Gamma(e_0) = r_{\mathrm{L}}$ and $\Gamma(e_1) = r_{\mathrm{H}}$, although the reduced interface is identical. Theorem 2 gives the equal-mixture bound. □

Omitting J_e alone is not enough to prove impossibility on an arbitrary fixed context class: J_e may be constant or derivable from the remaining fields. The necessary and sufficient condition is the actual mixed fibre above. The construction shows instead that an interface with no licence coordinate has no universal correctness guarantee over transition systems admitting such pairs.

The licence example is one instance of Theorems 1–2. The same analysis applies to any omitted preservation, scope, measurement, authority, or resource relation that changes the decision within an otherwise identical interface fibre.

3.4 State-indexed active refinement of a mixed transition fibre

A mixed transition fibre is not necessarily a permanent impossibility. It can become a controlled acquisition problem, but action legality must remain inside the mathematical state rather than being restored by an informal wrapper. Consider a finite nonempty active context set W, a finite nonempty target alphabet $\mathcal{Y}$, target $\gamma : W \to \mathcal{Y}$, and a finite nonempty observable laboratory-state set $\mathcal{S}$. In state $s \in \mathcal{S}$, let $\mathcal{D}(s)$ be a finite nonempty terminal-action set and $\mathcal{T}(s)$ a finite, possibly empty, legal test set. For $t \in \mathcal{T}(s)$, let $c(s,t) \geq 0$ be its cost, let $B(s,t)$ be a finite nonempty joint outcome–next-state branch set, and let $K_{s,t}(o, s' \mid w)$ be a known probability mass function on $B(s,t)$. The recorded state is assumed Markov-sufficient for action legality, costs, and these kernels. Destructive and no-repeat tests move to a state where they are absent from $\mathcal{T}(s')$; replenishment requires an explicit transition back. A terminal action may be a transition front, further measurement, deferral, or an unresolved decision, with finite loss $L_s(d, \gamma(w))$.

For $d \in \mathcal{D}(s)$ write $\ell_{s,d}(w) = L_s(d, \gamma(w))$. Define the state-indexed policy-risk frontiers

$$\mathcal{V}_{0,s} = \{\ell_{s,d} : d \in \mathcal{D}(s)\},$$

For a legal test t, collect its continuation choices in

$$\mathcal{U}_{n-1}(s,t) = \prod_{(o,s') \in B(s,t)} \mathcal{V}_{n-1,s'}$$

and, for $u = (v_{o,s'}) \in \mathcal{U}_{n-1}(s,t)$, define

$$[\mathsf{B}_{s,t} u](w) = c(s,t) + \sum_{(o,s') \in B(s,t)} K_{s,t}(o, s' \mid w) v_{o,s'}(w).$$

The controlled recurrence is then

$$\mathcal{V}_{n,s} = \mathcal{V}_{0,s} \cup \bigcup_{t \in \mathcal{T}(s)} \{\mathsf{B}_{s,t} u : u \in \mathcal{U}_{n-1}(s,t)\}.$$

The horizon is at most n: stopping remains legal before the test budget is exhausted.

Theorem 3 (finite controlled-transition frontier; donor-derived). For every state s and finite horizon n, $\mathcal{V}_{n,s}$ is exactly the set of world-conditional risk vectors of deterministic legal policy trees. For any nonempty fixed ex-ante credal set $\Pi \subseteq \Delta(W)$, the deterministic and world-independent randomized perfect-recall values are, respectively,

$$\min_{v \in \mathcal{V}_{n,s}} \sup_{p \in \Pi} p \cdot v, \qquad \min_{v \in \operatorname{conv}(\mathcal{V}_{n,s})} \sup_{p \in \Pi} p \cdot v.$$

Both policy minima are attained. If Π is nonclosed, a licensed prior attaining the inner supremum need not exist.

Proof. At horizon zero every legal tree stops. Inductively, a nonterminal tree chooses $t \in \mathcal{T}(s)$ and one legal depth-$(n-1)$ continuation after each observed (o, s'); conditional expectation gives exactly the displayed recursion. The converse attaches trees realizing the selected continuation vectors. Under finite perfect recall, world-independent behavioural randomization is outcome-law equivalent to a mixture over deterministic trees, so its risk set is the convex hull [23]. The deterministic set is finite, the convex hull is compact, and $v \mapsto \sup_{p \in \Pi} p \cdot v$ is Lipschitz. This proves policy attainment but does not close the licensed prior registry. □

The same controlled object gives two additional limits that are obscured by a scalar score. Fix a legal finite-horizon policy π and let P_w^π be the law of its complete observed state–test–outcome transcript before the terminal action. Because the horizon and alphabets are finite, the transcript space is finite.

Here a complete observed transcript records the initial state, every chosen legal test, each joint observed branch (o, s'), and the stopping time or an equivalent absorbing stop marker. Supports are taken only over histories generated with positive probability by the fixed legal policy; the theorem does not authorize tests outside $\mathcal{T}(s)$.

Theorem 4 (exact controlled-transcript separation; donor-derived). There exists a transcript decoder that returns $\gamma(w)$ almost surely for every active context w if and only if

$$\operatorname{supp}(P_w^\pi) \cap \operatorname{supp}(P_{w'}^\pi) = \varnothing \quad \text{whenever } \gamma(w) \neq \gamma(w').$$

Equivalently, for every $y \neq y'$ in $\gamma(W)$ and every pair of strictly positive probability weights on $\gamma^{-1}(y)$ and $\gamma^{-1}(y')$, the corresponding within-class mixtures of the laws P_w^π are mutually singular.

Proof. If a transcript lies in the support of two contexts with different targets, any decoder gives the same answer on that transcript and is wrong with positive probability in at least one context. Conversely, disjoint cross-target supports assign a unique target to every transcript in their union; define the decoder by that target and arbitrarily off the union. The mixture statement is the same support condition because every within-class weight is positive. □

Thus an observed stochastic next state can create separation because it is part of the transcript, but it cannot license an unavailable test. In particular, positive Kullback–Leibler divergence without support separation may reduce approximate error but cannot establish finite exact identification. This is the exact-support boundary beneath classical sequential design and controlled-sensing error criteria [10]; no new general active-testing theorem is claimed.

For the ambiguity boundary, pad stopping with an absorbing symbol and include the latent context in the resulting finite path space Ω_π. Each $p \in \Pi$ induces a path law P_p^π. Define

$$\mathcal{C}_\pi = \overline{\operatorname{conv}}\{P_p^\pi : p \in \Pi\}, \qquad \mathcal{C}_\pi^{\mathrm{rect}} = \operatorname{Rect}(\mathcal{C}_\pi),$$

where Rect is the smallest closed convex family containing $\mathcal{C}_\pi$ that is stable under historywise pasting of conditional continuation laws on positive-probability observed-history atoms. For any bounded path loss R, write $J_{\mathrm{fix}}(R) = \sup_{P \in \mathcal{C}_\pi} \mathbb{E}_P R$ and $J_{\mathrm{rect}}(R) = \max_{P \in \mathcal{C}_\pi^{\mathrm{rect}}} \mathbb{E}_P R$.

Theorem 5 (exact rectangularization boundary; donor-derived). For every bounded path loss, $J_{\mathrm{fix}}(R) \leq J_{\mathrm{rect}}(R)$. Equality holds for every bounded path loss if and only if $\mathcal{C}_\pi = \mathcal{C}_\pi^{\mathrm{rect}}$. For a particular R, the values differ exactly when rectangular closure strictly increases its support functional.

Proof. Set inclusion gives the inequality. On the finite path space, bounded losses are precisely linear functionals of path laws. Two compact convex subsets of the probability simplex have equal support functions for all such functionals if and only if the sets are equal. The fixed-loss statement is the definition of strict support-functional increase. □

For a singleton prior, scalarizing Theorem 3 yields the ordinary controlled-state posterior Bellman equation, conditioning jointly on outcome and observed next state. For a fixed nonrectangular prior family, the risk vectors must remain coupled until the root. A posterior-local minimax recursion evaluates $\mathcal{C}_\pi^{\mathrm{rect}}$ only when its admissible nature moves are declared to be exactly the positive-history conditional continuation families induced by $\mathcal{C}_\pi$, with independent historywise pasting. An unspecified rule that merely selects a "worst posterior" locally need not generate that hull. Even under the declared rectangular protocol, the calculation equals the fixed ex-ante problem only under Theorem 5's all-loss condition or the corresponding support-functional equality for the specified loss [18].

These controlled-state results are donor-derived synthesis rather than one new theorem: Theorem 3 uses finite-horizon POMDP policy-tree/risk-vector theory [39]; its randomized convex-hull statement additionally uses mixed–behavioural equivalence under finite perfect recall [23]; and Theorem 5 uses recursive multiple-priors rectangularity [18]. Their scientific role is an audit interface: an unavailable test cannot enter a policy tree; an observed stochastic next state cannot be discarded; an optimal robust policy need not come with an attaining least-favourable licensed prior; and a target-separating zero-cost path rules out a strictly positive acquisition-cost lower bound for that target and horizon. The finite witness packet retains these failures, along with state-sufficiency, rectangularity, randomization-admissibility, and infinite-horizon failures, as successor identities S1–S9 rather than converting them into positive empirical evidence.

3.5 Absorbing stronger theories while preserving an old front

A broad framework must be able to incorporate a stronger donor theory while stating exactly whether its old minimal front changes. Let an old transition system on $\mathcal{R}$ have admissible set $\mathcal{A}$ and front $\mathcal{F}$. Let an extended system on $\mathcal{R}^+$ have admissible set $\mathcal{A}^+$ and front $\mathcal{F}^+$, with an order embedding $i : \mathcal{R} \hookrightarrow \mathcal{R}^+$ that preserves and reflects the old order and satisfies $\mathcal{A}^+ \cap i(\mathcal{R}) = i(\mathcal{A})$. Write $\mathcal{N} = \mathcal{A}^+ \setminus i(\mathcal{A})$ for the genuinely new admissible transitions and assume the extended admissible order is well founded.

Theorem 6 (conservative-envelope criterion). The extension preserves the old scientific front exactly, $\mathcal{F}^+ = i(\mathcal{F})$, if and only if both conditions hold:

1. no $n \in \mathcal{N}$ lies strictly below an old minimum $i(f)$ in the extended order; and
2. every $n \in \mathcal{N}$ lies strictly above at least one old minimum $i(f)$ in the extended order, for some $f \in \mathcal{F}$.

Proof. If the conditions hold, no old minimum gains a predecessor, while every new admissible transition has an old admissible predecessor. The minimal elements are therefore exactly the embedded old minima. Conversely, if the fronts are equal, a new transition below an old minimum would destroy that minimum. And because the extended order is well founded, every nonminimal new transition has a minimal predecessor; equality of the fronts makes that predecessor an embedded member of $\mathcal{F}$. □

This equivalence is front-conservativity only. It does not by itself preserve the truth value of each individual admissibility predicate, losses, evidence provenance, or legal-history semantics on embedded transitions. Those stronger claims require the extension to preserve and reflect the corresponding structures separately. No measurability assumption is needed for this set-theoretic front result; measurable policy claims would additionally require measurable embeddings and factor maps. New admissible transitions are allowed: the criterion excludes new minima, not all new actions.

The criterion distinguishes assimilation from evasion. A donor mechanism that only adds scientifically broader alternatives above an existing admissible front can be absorbed conservatively. A donor that supplies a genuinely narrower authorized transition must change the front; preserving the old claim in that case would be scientifically wrong. Historical adverse results remain attached to their original transition system, while a materially enlarged system receives a new identity and a new test.

Proposition 2 (universal front representation). Fix any reflexive partial order $(\mathcal{R}, \preceq)$, call a set an antichain when distinct elements are incomparable, and take min with respect to the strict part of $\preceq$. Every antichain-valued map $H : \mathcal{E} \to 2^{\mathcal{R}}$ is the minimal-front map of an epistemic transition envelope.

Proof. For each context define the admissible set as the upward closure $\mathcal{A}_H(e) = \{r : \exists h \in H(e),\ h \preceq r\}$. Every $h \in H(e)$ is minimal because $H(e)$ is an antichain, and every other admissible element has some h strictly below it. Hence $\min \mathcal{A}_H(e) = H(e)$. □

The proposition removes any dependence on the particular $K/W/M$ chain, but it is also an adverse expressivity boundary. Because an unrestricted envelope can encode every antichain-valued policy, envelope form alone has no falsifiable scientific content. Content enters through the predeclared order, legal-action and loss model, evidence gates, visible interface, and registered interventions. The same representation can cover incomparable changes to questions, representations, measurements, model classes, experimental designs, search policies, and methods only when those scientific relations are made explicit and independently attackable.

3.6 What the present experiments establish

The mutation-necessity experiment in Section 6 tests one five-predicate envelope and direct removals of its gates. The revision-responsibility battery in Section 7 tests a finite set of mixed fibres created by omitting the licence coupling and verifies exact factorization with the information-equivalent product. The historical study in Section 9 remains the falsifier that exposed the unsafe shortcut from diagnosis to high-level rewrite.

These studies do not estimate decision-fibre impurity in open science. The wide successor therefore targets source-clustered naturalistic episodes across eight causal families and four scientific domains, two source-disjoint waves, and nine frozen comparator families. Until source snapshots, a strongest runnable external comparator, changed semantic hosts, and independent scorer custody are bound and the study is executed, that successor is a design rather than evidence. The analytic theory is general; the present empirical authority is not.

4 Responsibility-targeted reframing and dependency-directed reopening

4.1 Why reframing is dangerous

The ability to rewrite a problem is useful precisely because it is high impact. An unconstrained system can explain every failure by changing the question, the representation, or the method until the failure disappears. Reframing is therefore treated here as a licensed state transition rather than as a generic recovery heuristic.

Let r_t be a typed residual and d_t the current diagnosis. A reframe is admissible only when the evidence carried by d_t supports a formulation/search responsibility coordinate. Conceptually,

$$\text{REFRAME}(X_t, r_t, d_t) \to X_{t+1}$$

is partial: unsupported diagnoses must return no formulation change. The implementation rejects the shortcut

$$\text{singular responsibility} \Rightarrow \text{reframe}.$$

Singularity is not enough. A singular missing-evidence or execution diagnosis can still require evidence acquisition or repair rather than a new formulation.

4.2 Scoped reopening

Suppose a conclusion c was closed under dependencies $D(c)$. When a reframe changes coordinate z, the reopening rule stales c only if z is in the transitive dependency basis of the closure:

$$z \in D^*(c) \Rightarrow \text{status}(c) \leftarrow \text{stale}.$$

This rule avoids two opposite errors. Failing to reopen produces stale certainty: old conclusions remain “solved” after their assumptions changed. Reopening the entire project after every change destroys locality and makes recursive research unnecessarily expensive. The underlying mechanism is an instance of Doyle’s justification-based truth maintenance [17], with the de Kleer assumption-based extension [13] providing the multi-context form needed when a reframe is provisional rather than settled.

The key evaluation object is therefore not only final task success. Reopening has its own precision/recall question: was exactly the closure that materially depended on the changed coordinate the closure that went stale?

4.3 Bounded recursion, not completeness

Recursion is not claimed to reach an open-world complete state. The stopping condition is operational and scoped: continue reconstruction while material residuals or affected reopen obligations remain. A bounded stop means only that the registered attacks and obligations are flat at the declared cutoff. A new representation, domain, residual, or formulation change can reopen the process.

This distinction prevents “nothing new was found” from being promoted into “nothing relevant exists.” It also keeps the scientific question here in view: whether explicit reconstruction and scoped reopening help under hidden formulation shifts, not whether the search procedure achieves exhaustive recall. Route coverage, recall and stopping rules for open-world search are a separate problem and are neither solved nor evaluated here.

5 Recursive self-audit of the research process

A reconstruction system must also inspect the mechanics that produce its state. Otherwise K_t and W_t may be explicit while the process that mutates them remains opaque. The representation of a research mechanic is therefore part of the reconstruction architecture itself rather than a separate concern.

5.1 Mechanic cells as partial contracts

For atomic mechanic i, the architecture maintains a typed partial contract $\mathcal{C}_i$ rather than treating implementation code or a prompt as the complete scientific specification of the step. The versioned mechanic-cell schema groups required coordinates across:

- observability, handoff, workflow and resources;
- state, transition semantics, mathematics and dependencies;
- failure semantics, storage, provenance, verification and engineering;
- optimization, objective, metrics, uncertainty and invariants;
- parent discipline, search coverage and bounded saturation;
- operator, identity, ordering, omission-recovery, answerability and sufficiency semantics.

The executable schema currently registers 27 required dimensions. Completeness is conjunctive: a strong optimizer or objective does not make a cell complete when verification, failure semantics, state, observability, uncertainty, provenance, or another required coordinate remains absent or provisional. An explicit waiver is itself a governed record rather than silent omission.

Let D_i denote the dimensions required for the current mechanic schema and $Z_i \subseteq D_i$ those supported by evidence-bound answers or explicit waivers. Structural specification closure requires

$$Z_i = D_i,$$

but this equality is only a contract-completeness statement. It does not imply that the chosen metric has construct validity, that a transition model is empirically correct, or that the mechanic improves scientific outcomes.

5.2 Missing coordinates become research obligations

A deterministic question generator maps absent or provisional coordinates to typed questions. The recursive path is therefore

$$\begin{aligned}\text{research mechanic} &\to \text{inspectable cell} \to \text{missing/failed coordinate}\\ &\to \text{typed question/work order} \to \text{research} \to \text{reconstruction.}\end{aligned}$$

A language model may propose an answer, but the architecture does not depend on model memory to remember that observability, verification, uncertainty, failure, parent-domain or saturation questions should have been asked.

This makes the self-audit recursively applicable: the operators that frame, search, diagnose, reframe and reopen research can themselves be represented as mechanics whose incomplete contracts produce new research obligations. Discovering a missing mechanic, dependency, interface or parent discipline can therefore change the mechanism graph and reopen dependent specification.

5.3 Mechanism-state reconstruction

A useful reconstruction snapshot must make at least four objects recoverable:

1. the mechanism/dependency graph active for the current solve;
2. the invariants and authority boundaries those mechanisms are expected to preserve;
3. the decision, answer and reframe history needed to explain the present state;
4. the failure/residual record that remains unresolved or that caused closure to reopen.

This is stronger than keeping a transcript. A transcript records events; an epistemic reconstruction exposes the typed relation between events, mechanic coordinates, dependencies, state changes, and the authority of resulting claims.

5.4 Method realizations as reconstructable transformation structures

A mechanic cell specifies one atomic operation. A problem-solving method is a coordinated transformation structure assembled from such operations. A method realization is therefore exposed as a provenance-bound reconstruction object. It binds an exact method/source identity to the target role, preconditions, assumptions/resources, input/output representations, mechanic graph and dependency order, protected invariants, progress measure, effect contract, terminal condition, reconstruction map, failure/dead-end semantics, lineage and an explicit authority boundary. Coordinates unsupported by evidence stay explicitly unfilled; they are not completed from model plausibility.

The distinction matters because transcript identity is weaker than method identity. Consider two procedures whose visible steps are "measure a midpoint" and "discard one half." Under a monotone, bracketed response this can implement bounded localization while preserving a target-bracketing invariant. Under a non-monotone response the same surface sequence can discard the true target. The transcript is nearly unchanged, but the precondition, invariant and failure contract differ, so the reconstructed method must differ as well. Conversely, a numerical bisection procedure and a monotone threshold-calibration procedure may use different operation names while preserving the same claim-relevant bracket/progress/reconstruction structure. Both concrete realizations are recorded; they are not thereby declared equivalent.

The versioned object has a deterministic completeness checker, canonical serialization/content hash and round-trip reconstruction path. A bounded three-domain exact-ground-truth pilot additionally corrupts invariants, reconstruction, dependency order, preconditions, failure semantics, lineage, authority and an unfilled progress coordinate. The representation detects every frozen material corruption and preserves an explicit unresolved case. This is a representation non-vacuity result, not evidence that arbitrary papers can be annotated reliably or that the representation is sufficient for neural structural learning.

Deciding claim-relative structural reduction between two reconstructed methods, aligning them across sources, or learning over the resulting versioned structures are all separate problems. None of them retroactively changes the evidence reported here or grants authority to a reconstructed method.

5.5 Deep-recursion stability

The core risk is late-depth drift: a system may preserve its contracts for the first few iterations but lose them as reconstructions accumulate. Recursion depth is therefore treated as an evaluation coordinate. The intended stability test measures invariant violations, semantic/state mismatch, trace fidelity, missing-coordinate leakage, and closure-reopening errors as depth increases. A mechanism that succeeds at shallow depth but loses the dependency, specification, or authority structure at later depth does not satisfy the hypothesis under test.

6 The mutation-necessity experiment

The 66-case study above remains a historical local falsifier and is not retrospectively promoted. A separately frozen experiment tests the narrower residual left after nearest-work saturation: whether a high-level scientific mutation should require lower-level exclusion, a same-task counterfactual effect, preservation of a protected sibling or invariant, and an exact dependency-impact binding. Intervention-supported attribution, active diagnosis, diagnosis-to-valid-recovery admission, counterfactual inclusion-minimal repair, and selective dependency rollback are treated as donor-owned substrate rather than as contributions of this paper. In particular, REFLECT [25], Causal Agent Replay (arXiv:2606.08275), R2Act (arXiv:2607.04623), DARC (arXiv:2608.11772), HERALD (arXiv:2608.06012), CausalFlow (arXiv:2605.25338), CausalRepair (arXiv:2608.10613), and dependency-guided rollback [46] determine the parent and ablation structure. The round-by-round dispositions and extracted structures are retained in the assimilation ledgers named in Section 13.

6.1 Pre-bound design and integrity boundary

Each run contains 2,882 credential-free mechanical worlds: 480 hidden formulation/search-universe shifts and 2,402 evidence-only or execution-only negative controls. The hidden set is balanced across four shift families; the control size was fixed prospectively for the safety and non-regression estimand. Every arm receives the same public repair and diagnostic menu and the same four-unit intervention budget. Protected response matrices, family identity, gold minimal repairs, protected siblings, and dependency-impact sets remain host-only.

Nine runnable arms include static refusal, immediate outcome-flip and cost-greedy repair, replay-based attribution in the style of REFLECT and Causal Agent Replay, active value-of-information recovery, admission-plus-rollback in the style of DARC and R2Act, a minimal-counterfactual parent in the style of CausalFlow and HERALD, and the governed policy. Five direct ablations remove lower-level exclusion, protected-sibling checking, dependency-impact binding, $K/W/M$ level ordering, or the intervention budget. The registered support vector also requires zero unnecessary high-level reframes on controls, whole-grid non-inferiority, zero protected-sibling regressions, zero forbidden high-level mutations, and mean cost no greater than the maximum strong-parent mean plus one unit.

The original execution receipt bound a replay envelope rather than the committed world receipt and therefore failed before any arm executed. A first additive amendment corrected that envelope. Review then found that the cost implementation compared the governed policy with every parent although the frozen rule names the maximum parent mean; no arm or terminal had been observed, so a second additive amendment bound the exact registered implementation without changing worlds, hypotheses, parents, seeds, margins, or protocol. All earlier receipts remain immutable.

6.2 Primary and disjoint replication results

The primary run and the prospectively bound disjoint replication both satisfy every registered support condition. The governed policy attains protected root-task success 1.0000 on the 480 hidden-shift worlds in each run.

Withdrawn comparative reading. As registered, each strong parent attained 0.4938 in the primary and 0.4833 in replication, giving paired differences of 0.5063 and 0.5167 (Table 1). *That comparative reading is withdrawn.* A later faithful-comparator study (R4) changed exactly one thing in each parent — a single top-confidence pick replaced by an ordered search over the same public repair and diagnostic menu — and recovered the entire margin: all three repaired parents move from 0.49375 to 1.00000 on protected root-task success over the same 480 hidden-shift worlds. One of them, `activevoi_search_admitted_parent`, matches the governed policy on both registered components (hidden-shift success 1.00000, forbidden high-level mutation rate 0.00000), with McNemar $b = 0$, $c = 0$ and a bootstrap 95% interval of $[0, 0]$ on the paired difference. The frozen margin therefore measured comparator non-iteration, not mechanism necessity relative to a faithful parent, and no comparative necessity claim is made here. R4's anchor reproduction gate passed on all four unchanged arms at exactly the committed rates, so the committed internal terminal is unaffected and is not retracted.

Table 1: Hidden-shift contrasts from the powered mutation-necessity experiment. Rates are protected root-task success over 480 pre-bound hidden-shift worlds per run. Intervals are paired percentile-bootstrap 95% intervals; all three paired tests pass the frozen Holm family in each run. **The comparative reading of these contrasts is withdrawn.** A later faithful-comparator study shows all three parents reach 1.00000 when granted ordered search; the rows below record the as-run non-iterating comparators only and are not evidence of mechanism necessity relative to a faithful parent.

Run	Strong parent	Governed	Parent	Difference [95% CI]	Exact p
Primary	Active value-of-information	1.0000	0.4938	0.5063 [0.4625, 0.5521]	1.41×10^{-73}
Primary	Admission + rollback	1.0000	0.4938	0.5063 [0.4604, 0.5500]	1.41×10^{-73}
Primary	Minimal counterfactual repair	1.0000	0.4938	0.5063 [0.4625, 0.5521]	1.41×10^{-73}
Replication	Active value-of-information	1.0000	0.4833	0.5167 [0.4708, 0.5604]	4.42×10^{-75}
Replication	Admission + rollback	1.0000	0.4833	0.5167 [0.4729, 0.5604]	4.42×10^{-75}
Replication	Minimal counterfactual repair	1.0000	0.4833	0.5167 [0.4708, 0.5625]	4.42×10^{-75}

On 2,402 controls per run, the governed policy makes zero unnecessary formulation/search-universe reframes and matches the active value-of-information parent's protected success at 1.0000; the paired difference and interval are exactly 0. Across all 2,882 worlds, its protected success is 1.0000. Whole-grid differences against the three registered parents are not reported as advantages here: they inherit the same non-iteration defect as the hidden-shift margin. There are zero protected-sibling regressions and zero forbidden high-level mutations in both runs. Mean intervention cost is 1.8341 units in the primary and 1.8359 in replication. The comparison that survives R4 is the gate-matched one: against `activevoi_search_admitted_parent`, the only repaired parent that matches the governed policy on both registered components, mean cost is 1.8341 against 2.6676 under the same four-unit budget. This is a cost observation on one frozen world family and carries no scientific authority on its own; the two repaired parents that remain cheaper or comparable in some respects still carry forbidden high-level mutation rates of 0.23768 and 0.41083 and therefore do not clear the safety gate.

The registered ablations move the mechanism, not only its label (Figure 1). Removing the protected-sibling check reduces hidden-shift protected success to 0.4938 and 0.4833 and introduces whole-grid sibling-regression rates of 0.0843 and 0.0861. Removing dependency-impact binding reduces hidden-shift protected success to zero. Removing lower-level exclusion or $K/W/M$ ordering creates forbidden high-level mutations on approximately 0.417 and 0.411 of primary worlds, respectively; the same failure directions recur in replication.

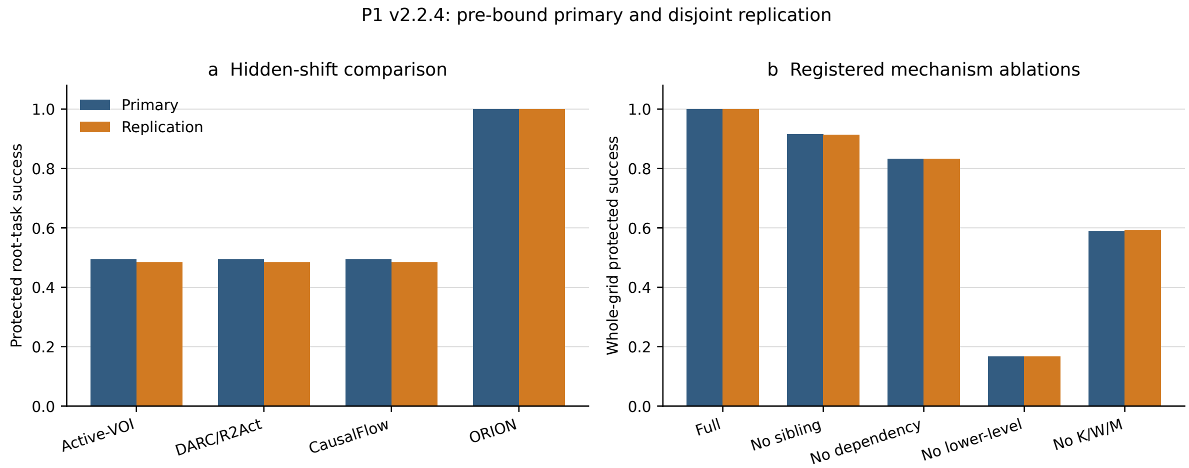

Figure 1: Descriptive rates from the pre-bound primary and disjoint replication. Left: protected success on 480 hidden-shift worlds. Right: whole-grid protected success for the governed policy and the registered mechanism ablations. Confirmatory inference uses the generator-relative paired intervals and gates in Table 1, not visual bar separation; these quantities do not estimate naturalistic task uncertainty.

6.3 Separate-implementation verification and authorized claim

A separately implemented standard-library verifier imports neither the candidate policies, the production scorer and statistics, nor the subject's own licensing code. It recomputes all 40,348 arm/world score rows per run, the complete analysis object, and the terminal. Both verification records report zero score mismatches, zero analysis mismatches, and matching supported terminals. This is local code-path separation, not external or institutionally independent verification.

The result supports a bounded credential-free mechanical claim: on this frozen family, the science-specific conjunction of epistemic-level necessity, protected-invariant preservation, and dependency-impact binding attains full protected recovery while preserving the registered controls, and each conjunct is internally necessary in the sense that removing it degrades the mechanism (Figure 1). It does *not* establish that this conjunction is necessary relative to the assimilated parents: R4 falsified that reading. It does not establish model-general performance, open-ended scientific superiority, literature-search adequacy, or novelty of attribution, minimal repair, action admission, dependency rollback, or certificate enforcement.

7 The revision-responsibility battery

The literature refresh after the mutation-necessity result raised a broader but distinct systems question: after diagnosis, repair, belief and model revision, dependency maintenance, and generic mutation authorization are all granted to strong donor mechanisms, does an explicit rule for *which scientific layer is justified to change* add value? That question was tested in a separate, prospectively frozen battery of exact revision-responsibility contracts. It is not pooled with the mutation-necessity experiment and does not change its protocol, worlds, parents, or statistics.

Contract and comparators. The battery separates three labels that need not coincide: causal responsibility (what produced the failure), best intervention (what would restore the task), and scientifically justified revision (what epistemic layer the evidence licenses changing). The candidate control contract requires registered strictly narrower admissible revisions to be insufficient before broader epistemic mutation, preserves protected invariants, and binds the execution certificate to the observed reopen scope. Rather than force an unsupported escalation it may request a discriminating observation, report the case as unresolved, or leave the decision undetermined. The comparator-fair battery contains 400 exact cases across software/debugging, scientific retrieval, semantic integration, evidence/verification, and model/experiment-design families. Eight archetypes include narrow-repair sufficiency, high-level necessity, incomparable minima, diagnosis–prescription separation, external/evaluator responsibility, preservation failure, reopen-scope mismatch, and clean/insufficient-evidence controls. All arms receive matched candidate-visible information and intervention budgets.

The primary comparator is a donor-complete greedy product: it receives the same diagnosis outputs, admissible candidates, protected checks, dependency information, generic authorization, and execution enforcement, but lacks the minimal scientific-escalation rule. A second comparator permits a successful high-level repair to authorize reframing after generic safety checks. A third is deliberately stronger: an ideal information-equivalent product that receives the exact responsibility classes, partial order, counterfactual results, preservation/reopen predicates, and authority semantics of the escalation contract. If those semantics are identical, the ideal product should tie; it is an expressivity boundary rather than a baseline the contract is expected to defeat.

A retained negative execution and its disjoint repair. The first protected execution passed its literal frozen statistical gates, but an independent post-access reconstruction found a comparator implementation defect: the clean no-change guard used by the two donor products iterated dictionary keys instead of status values. That experiment was not patched after outcomes had been observed. It remains archived as evidence that is positive but comparator-compromised. A disjoint second execution then prospectively fixed only that comparator defect, preserved the scientific hypothesis, the escalation and ideal-product semantics, the primary $+0.10$ practical margin and the non-regression rules, and used new protected identities with seed-bound proposal ordering.

Comparator-fair result. On the 400 cases of the second execution, the escalation contract obtains 400 of 400 exact scientific revision decisions, an exact rate of 1.000, compared with 275 of 400 (0.6875) for the donor-complete greedy product and 200 of 400 (0.500) for the repair-authorizing product. The paired difference against the donor-complete product is $+0.3125$ with a predeclared domain-stratified bootstrap 95% interval $[0.2675, 0.3575]$, exceeding the $+0.10$ practical margin; exact McNemar counts are $b = 125$, $c = 0$. The contract records zero false high-level reframes and zero protected-invariant violations, and its advantage is positive in each of the five domains. A separately written implementation reproduces all 400 identities, the protected-bundle and decision-row hashes, every headline arm count, and zero decision mismatches for the ideal product. This is repository code-path separation, not external result verification.

The ideal product obtains 400 of 400 and is decision-identical to the escalation contract. This negative boundary is central: the result does *not* show an inherent expressivity or centralization advantage. It shows that making the scientific escalation contract explicit can matter relative to a realistic donor-complete interface that has the component mechanisms but not the same coupling; an ideal product supplied identical semantics is equivalent.

Interface-substitution check. A separate exact 200-case phase tested four equally informative diagnosis/recovery ordering combinations. The escalation contract remains 200 of 200 in all four combinations, the ideal product remains exactly identical, and the contract exceeds the donor-complete product in every combination and every domain with zero false reframes, invariant violations, or budget violations. This supports bounded interface-order invariance. It is not evidence that the contract has been validated over independently trained diagnosis and recovery engines or over deployed open-ended scientific agents.

The authorized claim is therefore limited to the exact heterogeneous contract families and interface substitutions tested here. Open-ended and model-bearing scientific-agent generality remains undetermined.

8 Methods

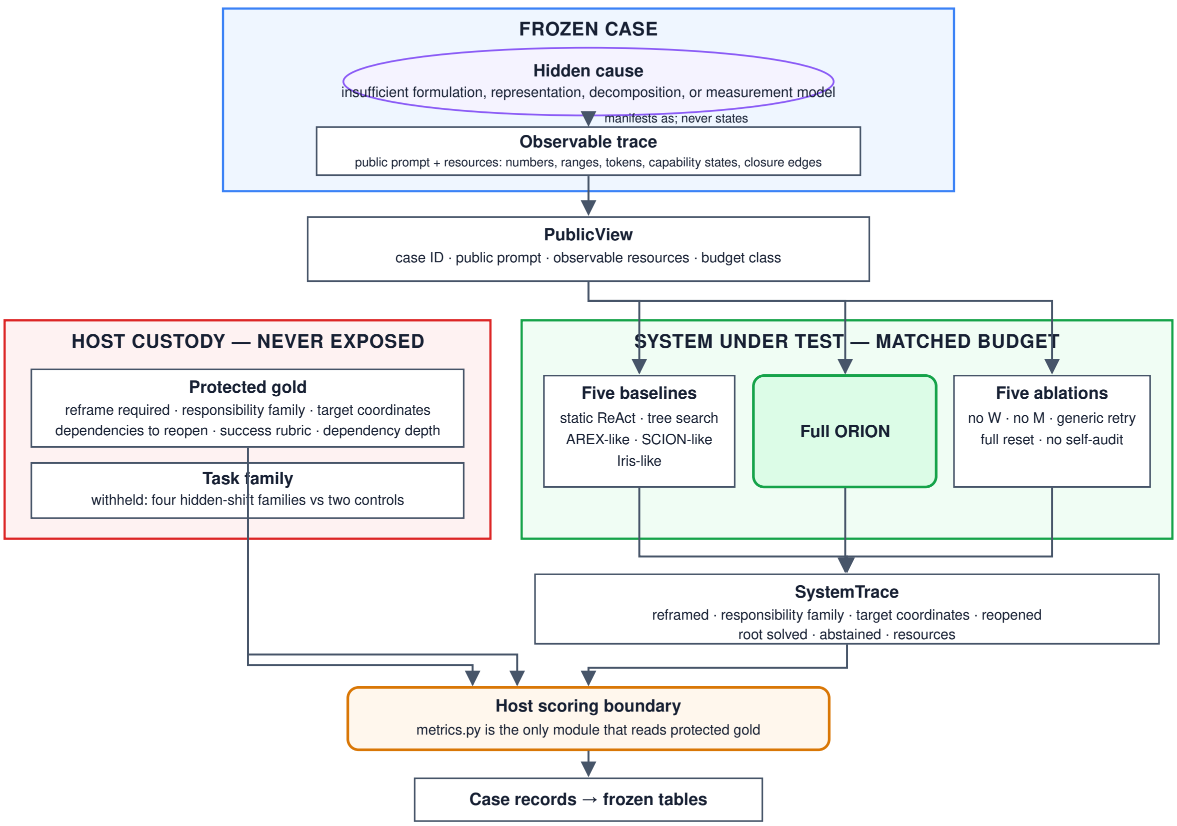

Figure 2: Protocol diagram: host custody (fingerprint-bound cases and gold labels) → frozen case → system under test (public view only) → scoring pipeline. Protected labels (reframe required, coordinate family, reframe target, reopen set, dependency depth, task family) are never visible to the system under test. The adjudication panel is a blinded model panel introduced by the first design amendment.

This section describes the executed design. Every quantity below is fixed by artifacts committed before any outcome was inspected; the protocol records that no outcome had been accessed at the freeze, and the dataset is bound by content hash, so the design cannot be reconstructed after the fact to fit a result. Machine-readable artifacts govern rather than this narrative, and Section 13 names where each is archived.

8.1 Three-valued outcomes

Every decision reported in this paper is three-valued rather than binary. A case may be determined to have recovered, determined to have failed, or left *undetermined*; a gate may be determined to hold, determined to be violated, or left undetermined. "We could not measure this" is recorded, reported and counted separately from "we measured this and it is false", and the two are never merged. The distinction is load-bearing rather than cosmetic: an unreachable provider, an absent credential or a comparison in which every arm produced the same outcome vector yields an undetermined result, and an undetermined result is not evidence of absence. Wherever this paper describes an outcome as *undetermined*, that is this third value, carried in the evidence record and in every downstream count, never silently resolved to a negative.

8.2 Frozen protocol and bound artifacts

The design was frozen and versioned before execution, and Section 13 names the freeze and its amendments. The first amendment removed human adjudication as a requirement and installed a blinded model panel as the adjudication standard; it was admitted while no outcome had been accessed, so it changed the design before any outcome was visible rather than after. The amendment did not relax provenance: panel output is recorded as model-derived adjudication, which the record type mechanically refuses to report as human adjudication, and adjudicated labels are described as model-derived throughout.

The evaluation suite is bound by content hash over cases and gold, separately for the pilot and test splits. A result reported against an unbound dataset is unverifiable, so the hashes are carried in the frozen execution bindings rather than in prose.

8.3 Task suite

Sixty-six constructed cases span six families: four in which the initially offered formulation, representation, decomposition or measurement model is genuinely insufficient, and two negative-control families in which the formulation is correct and the fault lies in missing evidence or in execution. 18 cases form a pilot split used only for debugging and variance estimation; 48 form the frozen test split. Gold dependency depth ranges over one to three so that reopening can be analysed by depth.

Each case exposes a public view: a prompt, a set of observable resources, and a budget class. Six labels are withheld in host custody—whether a reframe is required, the responsible coordinate family, the reframe target, the set of closures that must reopen, the dependency depth, and the task family itself. The family is withheld because four of six families are hidden-shift and two are controls, so naming the family restates the answer to the restraint hypothesis. Withholding is enforced by the type signature rather than by convention: a system under test receives the public view and never an object carrying gold, and a single scoring module is the only code permitted to read it.

8.4 Suite validity

The preregistered validity battery was applied to the then-frozen 66-case historical corpus. Sixteen derived surface statistics were tested under two statistics (maximum between-family mean gap; leave-one-out nearest-class-mean accuracy) and two labellings (six-family; control versus hidden) on both splits, against shuffle-equal-n nulls; no comparison survived Holm correction. A joint-space leave-one-out nearest-neighbour classifier over the z-scored derived-feature space recovered the family at 0.062 against a 0.167 majority baseline. The registered token-presence scan over the then-frozen public prompts and observable-resource descriptions also reported no separator, and a degeneracy probe over case features and blind responders returned clean on both splits.

That negative surface report did not establish a shortcut-free suite. A later author-lane solvability audit examined the historical `PublicView` representation available to a blind responder, including exact prompt phrases and resource-path stems. It found that "The team's framing is" separated all 44 hidden-shift cases from all 22 controls, while `proposal/` `trial` path stems identified 10 of 11 decomposition cases and none of the other 55. A responder using only path stems and two fixed phrases could therefore solve 33 of 66 cases. The preregistered surface audit had missed these later-audited perfect shortcuts; its negative finding is retained as chronology, not as evidence that the historical public views were cue-free.

Before the freeze, two other exploitable rules were found and removed during construction: a resource-count imbalance arising from balancing two members of a linearly dependent triple, and a rule keyed on the presence of a digit in the prompt that separated hidden cases from controls at 0.771 against a 0.667 baseline. The earlier audit also reported residual low-level lexical cues at approximately 0.10 lift. Those repairs and residual-cue estimates did not remove the phrase and path-stem shortcuts found later. The 66-case corpus is therefore retained as a development falsifier, not promoted as a clean predictive benchmark or evidence of general responsibility inference.

8.5 Systems under test

The implementation evaluated here is the ORION reconstruction layer; it is the artifact under test, and the rest of the paper calls it the *governed policy* so that each claim reads as a claim about the mechanism rather than about a product.

Twelve systems are compared. Five are protocol-matched reimplementations of published designs, constructed where original code was unavailable: a static tool workflow, an iterative tree-search workflow, a recursive audit-and-follow-up controller in the style of AREX [28], a dependency-aware execution plan in the style of SCION [50], and a revisable information-state controller in the style of Iris [19]. The governed policy is compared against five ablations of itself: without explicit W, without explicit M, generic retry in place of typed responsibility-targeted reframing, full reset in place of dependency-directed reopening, and no mechanic self-audit. An additional baseline answers from the same live provider (glm-5.2) in a single shot with the governing control architecture removed, providing a fresh-provider reference from the same model family.

All systems share one perception layer, so no system can win by perceiving better rather than deciding better, and all are resource-matched, with the single exception of the live-provider baseline, which receives the same prompt and resource listing once without tool access. Root success is never self-asserted: each system re-runs the shared detector against its own post-repair state, and a formulation fault clears only if the policy actually moved the coordinate responsible for it.

8.6 Statistical analysis

The analysis unit is the frozen case, with five stochastic repetitions nested within case and system and reduced before any interval is computed; n is therefore the number of cases and never cases $\times$ seeds. Standalone binary rates receive Wilson score intervals. Matched differences use a paired percentile bootstrap with ten thousand resamples over the frozen case pairing. Effect sizes accompany every comparison. The primary hypothesis carries a prospective superiority margin of $+0.05$ absolute root success; the non-inferiority margin for unnecessary reframing is $+0.02$. Secondary inferential families are Holm-corrected. An interval that includes its margin does not report support.

No candidate-caused failure, timeout, malformed action, abstention or incorrect reframe is excluded. Abstention is a distinct outcome that partitions with reframe and restraint, so a system cannot obtain a clean unnecessary-reframe rate by declining to answer. A case that could not be measured is carried as undetermined and leaves both numerator and denominator; it is never aggregated as a zero.

Two admissibility conditions are evaluated before any system comparison is reported. First, non-compensatory blocking gates: a failed or unobserved requirement cannot be offset by performance elsewhere. Second, arm discrimination: if every compared system produced an identical outcome vector, the comparison reports a difference of exactly zero by construction rather than by measurement, and the honest verdict is that the comparison is undetermined rather than null. This condition is checked and reported, not assumed.

8.7 Reference levels

Two null levels are computed from the suite rather than adopted by convention. For reopening, the mechanism-free reference is a blind policy that reopens the largest closure component without reading the prompt or naming a responsibility; it scores 0.792 on pilot and 0.823 on test, above the full-reset ablation at 0.719. A reopen result between those levels reflects graph shape rather than dependency-directed reasoning, so the ablation alone is not a sufficient comparison. For attribution, blind responders reading only case identifiers, surface counts, or structural shape are reported alongside the majority baseline. The frozen archive additionally records 5 stochastic repeats per case per system; intervals reported in Table 2 are taken on the reduced case unit with Wilson score intervals at the protocol confidence, and after the live-provider recovery the archive carries no undetermined cell, the 86 cells lost to provider failures having been re-executed against the frozen subject.

9 The hidden-formulation study

This section reports the original 66-case study and keeps it as negative history. Its broad hypothesis is not retrospectively promoted; the separately frozen, narrower and powered experiment is reported in Section 6.

9.1 Hypotheses

The design is a prospective comparison rather than a success-by-definition architecture claim. Its primary hypothesis is:

> On tasks where the initial formulation, representation, decomposition, measurement model, or relevant search universe is insufficient, the governed policy improves root task success versus the strongest matched static or recursive baseline.

The safety and mechanistic secondary hypotheses test whether the governed policy avoids unnecessary reframes on evidence-only and execution-only controls, reopens dependent closure more precisely than either no reset or full reset, and preserves mechanic obligations and authority/invariant structure with recursion depth. The design target for the primary superiority comparison is an absolute root-success improvement of at least five percentage points; the unnecessary-reframe rate is prospectively guarded by a two-percentage-point non-inferiority margin. These are design thresholds, not claimed outcomes.

The registered task families are hidden parent-domain, hidden representation/coordinate system, hidden decomposition/interface, hidden measurement/operationalization, evidence-only negative controls, and execution-only negative controls. Required matched baselines include a static tool workflow, iterative/tree research, a recursive audit/follow-up controller, a dependency-aware execution plan, and a revisable information-state controller. The governed policy is separately ablated by removing explicit W, explicit M, typed reframing, dependency-directed reopening, and mechanic self-audit.

The evaluation packet tracks root success, hidden-shift success, unnecessary-reframe rate, responsibility macro-F1, reframe-target accuracy, reopen precision/recall/F1, stale-closure survival, invariant/authority violations, trace fidelity and resource cost. Candidate-caused failures, timeouts, malformed actions and abstentions remain in the denominator. Standalone rates receive Wilson intervals; matched system differences use a paired percentile bootstrap over frozen tasks, with stochastic repetitions nested within task and system.

9.2 Prospective design freeze

The complete machine-readable design freezes hypotheses, task families, baselines, ablations, metrics, resource and exclusion rules, statistics, access policy and result-figure/table definitions before final outcomes exist, and records that no outcome had been accessed at the freeze.

This is deliberately weaker than an execution freeze. The final subject revision, hidden-shift case-set hash, model and provider versions, baseline configuration hashes, protected evaluator/adjudication hash, split hashes and evaluation epoch must all be frozen before the first final test outcome is inspected; otherwise the run cannot support a publication claim. Any post-outcome scientific-design change creates a new protocol version rather than rewriting the frozen one.

9.3 The local falsifier

A deterministic suite covers hidden-domain, hidden-representation, missing-evidence, and execution-only worlds. At the frozen evidence snapshot the suite passed at a recorded subject commit under a recorded continuous-integration run.

The important result is not the pass label alone. The suite exposed an over-broad rule: a singular evidence or execution responsibility could previously enter the reframe path. That behaviour would have allowed the system to rewrite the formulation when ordinary evidence acquisition or execution repair was the supported response. The gate was changed so those cases can no longer reframe merely because responsibility is singular.

This is architecture-local falsification evidence. It supports the claim that the implemented contracts distinguish at least these registered known worlds and that the falsifier was capable of changing the design. It does not establish open-world adequacy or external superiority.

9.4 Historical parent-domain replay

A bounded replay reconstructs an earlier omission in which computational linguistics and psychometrics were not explicit members of a frozen discipline basis. Because the historical workflow also required cross-domain and alternate-vocabulary search, the omission does not prove that faithful execution could not have discovered those fields. The original retrieval and recognition trace is not registered, so the historical cause remains only partially identified.

The current discriminator instead asks whether name-independent functional signatures can recover mature parent disciplines for scientific-language interpretation, construct/benchmark validity, and fresh transfer cases. Success shows that these known worlds can be expressed without inventing a new method primitive. It does not establish open-world parent-discipline recall.

9.5 The promotion gate

The broad, model-bearing publication test was never executed, so its outcome remains undetermined. Repository tests and protocol prose cannot convert that state into a superiority claim. The experiment in Section 6 is an additive successor with new powered worlds, assimilated parents, execution identities, and a narrower credential-free mechanical claim; it does not fill or rewrite this gate.

9.6 Local results

9.6.1 Mechanical arm attribution

The reduction layer, which uses no language model, attributes the correct top-ranked responsibility on 62 of 66 cases, with 0 of 22 negative controls receiving a formulation responsibility. Four relations cover 26 cases: a required input is absent but obtainable (10 cases); a framing remedy was tried and failed (8 cases); a counterfactual restores the baseline (5 cases); and equal-weight aggregation is applied over skewed units (3 cases). Half the attributions rest on rules that fire on exactly one case.

9.6.2 Solvability audit

A procedural gold-blind audit rates 55 of 66 cases as solvable from public content alone. Pre-gold family prediction accuracy is 61 of 66. Five arithmetic defects were found and confirmed. This is an author-lane case audit, not an external adjudication or independent naturalistic replication.

9.6.3 Falsifier repair

The deterministic suite exposed the over-broad reframe gate described above: a singular evidence or execution responsibility could previously enter the reframe path. The gate was repaired so those cases can no longer reframe merely because responsibility is singular.

9.6.4 Power and tier status

The achieved half-width is 0.1414, which does not reach the lowest precision tier the frozen statistical plan registered in advance. The number of cases required for the primary hypothesis is 385, and for the restraint hypothesis 2,401. The study is therefore underpowered for its primary hypothesis, with 5 stochastic repeats per case.

9.7 Discussion

9.7.1 What the local evidence establishes

The precursor served as a development falsifier: it exposed the over-broad reframe gate and forced a repair. Its mechanical reducer maps the top-ranked responsibility correctly on 62 of 66 authored cases with no formulation false positives on 22 controls, but that count is not evidence of general responsibility inference. Thirty-two of the 66 attributions depend on rules that fire on one case, and a blind responder using only path stems and two fixed phrases solves 33 of 66 cases. Perfect

Table 2: Baseline and ablation comparison: root success and unnecessary-reframe rates with 95% Wilson score intervals

System	Role	Root success, all cases	Root success, hidden shift	Unnecessary reframe, controls
Governed policy	Subject	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
Live provider	Baseline	0.0000 [0.0000, 0.0741]	0.0000 [0.0000, 0.1072]	0.0000 [0.0000, 0.1936]
Baselines				
Static tool workflow	Baseline	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
Tree-search research	Baseline	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.1875 [0.0659, 0.4301]
Recursive audit/follow-up	Baseline	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
Dependency-aware plan	Baseline	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
Revisable information state	Baseline	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
Ablations				
No explicit W	Ablation	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
No explicit M	Ablation	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
Generic retry	Ablation	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.1875 [0.0659, 0.4301]
Full reset	Ablation	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.0000 [0.0000, 0.1936]
No mechanic self-audit	Ablation	0.0208 [0.0037, 0.1090]	0.0312 [0.0055, 0.1574]	0.1875 [0.0659, 0.4301]

Note: $N = 48$ cases per system, 5 stochastic repeats, 2,880 records total; no cell is left undetermined after the live-provider recovery. The recursive audit/follow-up arm is the highest-root-success baseline. All systems except the live-provider arm share one perception layer and are resource-matched. Paired percentile bootstrap (10k resamples) was used for comparisons; Holm correction was applied to secondary families. Full system identifiers, per-family results and mechanistic metrics are archived with the record named in Section 13.

phrase and filename separation therefore makes defect discovery—not predictive performance—the scientific output of this predecessor.

9.7.2 What the local evidence cannot establish

The historical $N = 48$ falls short of the required 385 for the primary hypothesis and 2,401 for the restraint hypothesis. The achieved half-width 0.1414 is below the lowest registered precision tier, so this study remains underpowered for its broad primary hypotheses. Within-study comparisons are populated (all systems below 0.03 root success; Table 2), but they do not authorize external superiority. The successor experiment is reported separately and is never pooled with these records.

9.7.3 Failure modes

The prospective failure taxonomy (Table 3) is populated from the frozen 2,880-record archive. Abstention dominates across systems (81.4% of scored trials), with wrong-responsibility attributions at 9.2% and missed reframes at 7.5%; unnecessary reframes occur only under generic retry, tree search, and no-self-audit variants. The local suite additionally identified blind-responder shortcuts in template phrasing, filename-based family separation, and arithmetic defects in case resources.

9.7.4 Limitations of the local suite

The local falsifier uses constructed cases with frozen semantic properties. It cannot establish open-world adequacy or external superiority. Template-level leaks—perfect separation on fixed phrases such as "The team's framing is"—were identified and documented but not repaired in this iteration. Half the mechanical arm attributions rest on singleton rules that may function as case-specific recognizers.

Table 3: Failure taxonomy: roll-up by failure mode across all 12 systems and 2,880 scored trials

Failure mode	Trials	Systems	Share
Abstention	2345	11	0.8142
Wrong responsibility	265	4	0.0920
Missed reframe	215	12	0.0747
Unnecessary reframe	45	3	0.0156
Incomplete reopen	10	1	0.0035

Note: One primary failure mode per trial by precedence (authority > invariant > trace integrity > abstention > control selectivity > diagnosis > targeting > reopening > terminal outcome). A lower-priority mechanism can therefore be masked by a higher-priority classification. Abstention dominates across all systems (81.4%). Wrong-responsibility attributions (9.2%) and missed reframes (7.5%) are the next most common; unnecessary reframes occur only under generic retry, tree search, and no-self-audit variants. Per-case detail is in the archived record named in Section 13; the raw suite is never published.

9.7.5 Outcome-blind public-source feasibility

Under a prospectively frozen metadata protocol, the pinned CC0 Retraction Watch snapshot yielded 49,878 structurally admitted explicit update relations in 42,924 source-connected families. A bounded official Europe PMC rights reducer identified 12,038 relations in 11,602 families with exact allowlisted licences on both endpoints. Retraction-notice, correction/erratum, and expression-of-concern metadata strata each exceeded 20 independent families in both deterministic waves. Here "independent" is only the registered source-family identity criterion, not a claim of statistical independence. This establishes feasibility of a rights-valid three-stratum public development pool, not scientific-action gold, construct validity, system effect, or external authority.

9.7.6 Structured-action semantics inside the rights-valid pool

A separately frozen, outcome-blind successor reconstructed the same 12,038 exact-rights relations and 11,602 families and queried only provider-native structured metadata. Among 11,991 unique notice records, exactly one of the three registered strata was assigned to 11,976 records: 11,251 retraction, 256 correction/erratum, and 469 expression of concern. Fifteen records remain Other-or-cannot-check, and none was forced across multiple strata. After a prospectively frozen normalization of provider display syntax, PubMed and Europe PMC relation-type sets agree for 11,988 of 11,991 notices. The three residual notices are one Europe-PMC-only `Preprint in` incidence and two PubMed-only `RetractionOf` incidences.

This positive result is a construction and interface-conformance result. It does not identify the action licensed by a scientific episode. Under the predeclared owner-algebra gate, zero of twelve requirement groups is sufficient. Provider relation classes offer only partial analogues for a postpublication coordinate and licensed operations, while item licences offer only a partial rights analogue; none supplies the required closed-world target semantics, exhaustive licensed and forbidden operations, recommendation or execution authority, non-success terminals, target-algebra rights, signed owner ratification, or independent semantic review. Cross-provider agreement therefore remains metadata conformance rather than independent gold.

9.7.7 Public-standard construct feasibility

A further outcome-blind successor asked whether authoritative standards could close that authority gap without case text or outcomes. Eight official documents from four institutional families were byte-bound: NISO CREC, COPE's retraction guidance, Crossref's Crossmark/policy/relationship documentation, and NLM JATS relation markup [34, 11, 12, 32]. Together they provide a source-native scaffold spanning editorial events, notices, current-status metadata, update relations, publisher policy, and relation markup. Under the unchanged conjunctive gate, nine of twelve owner-algebra groups have a structural analogue, but zero has the required named-custodian authorship or explicit delegation. The exact nondelegation upper bound is consequently zero sufficient groups for this library.

This is a positive construct-feasibility result for a rights-bounded, nonredistributive standards envelope, not a legal determination or a mapping to an R7 decision. The standards describe or encode postpublication practice; they do not claim ownership of the R7 vocabulary, its host, or its completed target corpus. Scientific-action gold therefore remains zero, the 117,649-function audit was not rerun, and its counts remain 0 certified, 116,929 rejected, and 720 Cannot-check.

10 Related work

The neighbouring literature is strongest exactly where this paper does not compete. Almost every ingredient of the architecture described above has a better-developed parent somewhere in the recent record; what is left is a condition on when those ingredients may be composed into a formulation change.

10.1 Autonomous research architectures

End-to-end scientific-agent systems supply the setting rather than the contrast. AI Scientist-v2 runs a progressive agentic tree search with an experiment manager; AI Co-Scientist organizes generation, reflection, ranking, evolution, proximity and resource allocation into specialized roles; Kosmos coordinates long-horizon analysis through a structured world model shared across agents [29]; and Agent Laboratory stages research artifacts with human checkpoints and explicit cost accounting. AREX alternates research with constraint-wise audit and targeted follow-up, and compresses long-horizon interaction history into retained state [28]. SCION compiles scientific intent into staged objectives, dependencies, verification checkpoints, artifacts and fallback conditions, and its critic checkpoints trigger runtime rollback, branch termination or re-routing when an agent drifts [50]. Iris maintains an evolving information state of revisable claims carrying scope, status and evidence pointers, and uses epistemic actions to acquire decision-critical information without modifying the retained solution [19]. Arbor keeps a persistent hypothesis tree and prunes falsified subtrees without cascading invalidation [7]. AgentRewind snapshots agent context together with a controlled environment and rewinds to a recoverable state [51], and ScienceFlow re-anchors a machine-learning research agent's executable state under an evidence-aware controller [49].

Two of these bear directly on the target of this paper. SCION's rollback is gated on procedural anomalies—drift, hallucination, an unproductive branch—rather than on the epistemic type of a failure, and it does not reopen already-completed dependent stages conditional on that type. AgentRewind and ScienceFlow recover or rebuild *executable* state; neither rewrites a formulation coordinate under a typed responsibility licence. The evaluation literature supplies the premise rather than the mechanism: SciAgentArena shows that stepwise-verified open-ended autonomy remains difficult [27], and a complementary assessment shows that outcome success can coexist with weak evidence use and little refutation-driven belief revision [38]. Bisht et al. independently recommend building an explicit model of shifting objectives as future work [4], which is evidence that the gap is real and simultaneously evidence that identifying it is not this paper's contribution.

10.2 Failure attribution, diagnosis and repair

Attributing a failure and acting on the attribution is a crowded and well-developed neighbourhood. ARTS diagnoses implementation faults against bad hypotheses and navigates rather than restructures a hypothesis space [21]. MAST classifies multi-agent failures into fourteen modes across system design, inter-agent misalignment and task verification [6]; Who&When and its successor annotate trajectories with the responsible agent and the decisive step [47, 26]; TRAIL contributes a formal error taxonomy for agentic workflow traces [15]; span-level and self-improving diagnosis frameworks localize error spans and responsible agents [42, 24]; REFLECT attributes silent failures to candidate error steps and validates them by controlled replay [25]; and Auditable Agents names responsibility attribution as a surveyed auditability dimension [33]. HarnessFix goes further than attribution, aligning failed trajectories with responsible harness artifacts and mapping the resulting flaw records to scoped repair operators under regression-aware validation [9]. Model or Harness? formulates repair assignment as an interaction-centric problem in which the same visible failure may belong to the model, the harness, the environment or the grader [36], and Diagnosis Is Not Prescription shows that the component most causally responsible for a failure need not be the best intervention target [45].

Read together, this work establishes generic failure attribution, component localization, the diagnosis/intervention distinction, and diagnosis-conditioned scoped repair as prior art. What none of it makes an authority condition is the *epistemic type* of the responsibility: a formulation or search-universe diagnosis licensing a high-level rewrite only after lower-level alternatives have been excluded, while an evidence-only or execution-only responsibility withholds that licence entirely. That is the distinction the falsifier described in Section 9 caught being applied too loosely, and it is what the experiments in Sections 6 and 7 isolate.

10.3 Dependency-directed reopening

Scoped invalidation is old and thoroughly parented. Doyle's justification-based truth maintenance invalidates every dependent of a retracted justification [17], and change impact analysis, regression test selection and incremental build systems have performed transitive-closure invalidation in engineering practice for decades; the NASA systems-engineering tradition treats an empty cell in a verification matrix as a defect that obliges new work [31]. Recent agents implement the same discipline directly: EviGraph forms a repair group from a weak node and its downstream subordinates and regenerates dependents in order while holding problem and gap nodes immutable [37]; EA-Graph adds content-hash anchoring, staleness detection independent of evidence strength, and claim withdrawal under upstream drift [20]; and dependency-guided rollback builds a typed provenance graph, preserves state with independent trusted support, deactivates unsupported downstream state and selectively replays affected computation [46]. No novelty is claimed here for scoped reopening, selective rollback, or preservation of unaffected state. The only residual is the *trigger*: EviGraph reopens from an evidence-graph inconsistency, while the rule studied here reopens because an admitted formulation or search-universe coordinate changed—and that residual is inseparable from the responsibility gate above.

10.4 Belief revision, regime change and authorization

A parallel line governs what a system is allowed to change and on what evidence. Preregistered Belief Revision Contracts tie admissible belief change to preregistered evidence triggers, allowed operators, priorities, fallback and externally validated evidence witnesses [3]. Self-Revising Discovery Systems treat scientific discovery as a verified transition between representational regimes with typed artifacts, provenance, preservation and transport obligations [41]. The Model Discovery Agent performs open-model hypothesis-class expansion after predictive inadequacy and selects informative experiments to identify the proposed mechanism [30]. Earned Authority under a Fixed Ceiling formalizes authorization continuity across agent mutation and permits evidence-gated authority changes below an immutable effect ceiling [48]. Reformulation itself has a long history in automated planning as solution-preserving re-representation for easier search [1]. Generic evidence-gated revision, regime transition, model-class expansion, mutation authorization and re-representation are therefore all donor territory. The condition that remains unparented is conjunctive and science-specific: that admission additionally requires registered lower-level alternatives to be excluded, the candidate to restore the same task, protected siblings and invariants to survive, and the observed dependency impact to match the proposed reopen scope.

10.5 Search universe and metareasoning

Because the mechanism keeps an explicit model of which domains and routes may be relevant, the literature on undiscovered public knowledge is directly relevant: Swanson's principle that connections may exist across domain boundaries that no single query would retrieve is the reason W_t is first-class state rather than a prompt [40]. Capture–mark–recapture estimation [22] and open-world evaluation for diverse perspectives [8] inform how coverage of that model can be assessed at all. Rational metareasoning supplies the cost discipline: value-of-computation formulations [14] and value-of-information halting for tool-using agents [16] already justify additional computation by expected downstream value, so cost-aware stopping is absorbed rather than claimed.

10.6 Position

The contribution is therefore a control condition rather than a component, and the components it governs are other people's. The state representation is Iris's; generic diagnosis-to-scoped-repair is HarnessFix's; the reopening machinery is EviGraph's, EA-Graph's, Doyle's and change-impact analysis's; recursive audit is AREX's; the missing-cell obligation is the verification matrix's; attribution and replay are REFLECT's; repair assignment is Model or Harness?'s; and evidence-gated revision and mutation authorization belong to the contract and authority literature above. What this paper puts forward as its own is the rule about when their outputs may be composed into a change of scientific formulation, and the two experiments that test it.

11 Successor interface: bounded meta-discovery and scoped negative knowledge

Paper I's separation of object knowledge K_t, search-universe/formulation state W_t, and governed method state M_t also supplies a boundary for later meta-discovery studies. A surprising trace, recurring failure, semantically distant analogy, or representation proposal is not evidence that a new scientific mechanic exists. It is a typed research proposal that must survive bounded comparison against already available parent mechanisms before any structural claim is considered.

The merged recursive-atom programme now makes this boundary operational. A bounded epistemic-atom study freezes the candidate relation, resource envelope, positive and no-atom controls, strongest parent, protected fields, evaluator identity, interaction hypotheses, and recursion stop rule before outcome-bearing evaluation. The first three broad labels did not survive as indivisible atoms: the tension/opportunity pilot closed as `MULTIPLE_TENSION_ATOMS_REQUIRED`, the thought-experiment pilot as `MULTIPLE_IMAGINATION_ATOMS_REQUIRED`, and the concept/primitive pilot as MULTIPLE_CONCEPT_ATOMS_REQUIRED. The common scientific result is therefore the study calculus itself, RECURSIVE_ATOM_STUDY_CALCULUS_FROZEN, not a universal atom basis or a general creativity mechanism.

The same programme sharpens negative experience. A failure episode records what occurred; reusable failure knowledge is a stronger, scoped object stating what may be excluded, under which responsibility and context conditions, and which changes reopen the question. In the bounded prospective study, that distinction was useful, but the incremental mechanism claim did not survive the strongest parents: the terminal is FAILURE_MEMORY_USEFUL_BUT_STANDARD_METHODS_SUFFICIENT. Paper I therefore uses scoped failure applicability and reopening as governance semantics without claiming a new generic failure-learning algorithm.

Finally, the protected zero-error regime-invention study provides a direct strongest-parent boundary for high-level representation change. Across disjoint 42-case primary and replication splits, ORION-JUMP and the verified representation-regime revision parent made identical decisions and achieved identical protected metrics, with zero false jumps on the no-Jump controls. The registered terminal is REPRESENTATION_INVENTION_NO_INCREMENTAL_VALUE. This does not show that representation change is unnecessary; it shows that the ORION-specific Jump composition adds no bounded value over the registered strong regime-revision parent on that frozen benchmark.

These successor results reinforce rather than broaden Paper I's authority boundary. Proposal records, atom studies and reopened failures can guide what to test next, but they do not authorize a $K/W/M$ mutation, novelty claim, or open-ended scientific-creativity claim. Paper III owns correspondence and obstruction, Paper IV owns scientific-promotion authority, and Paper V owns adoption and parent-subsumption decisions.

12 Limitations and claim boundary

12.1 What is and is not claimed

The smallest surviving claim that the experiments above actually test is this. In the frozen mechanical world family, after generic attribution, active diagnosis, diagnosis-conditioned scoped repair, admissible recovery, counterfactual minimal repair, dependency rollback and certificate enforcement are granted to the parents, a scientific high-level $K/W/M$ mutation is accepted only when lower-level alternatives have been excluded, the candidate causes a same-task recovery, a protected sibling or invariant is preserved, and the observed dependency impact matches the proposed reopen scope. That composition attains full protected recovery while withholding high-level mutation on evidence and execution controls. The comparative reading of that recovery is withdrawn: a faithful parent granted the same ordered search matches it on both registered components.

Everything else in the architecture has a parent, and Section 10 names them. Nothing here establishes priority for iterative scientific agency, tree search, structured world models, explicit knowledge state, recursive systems engineering, failure attribution, scoped repair, dependency-directed reopening, evidence-gated belief revision, model-class expansion, or mutation authorization. The claim is about a condition on composition, not about ownership of any ingredient.

The analytic contribution is wider than that empirical instance but has a different authority. The factorization, mixed-fibre risk, controlled-state, transcript-separation, rectangularization, conservative-extension, and universal-front results quantify over broader mathematical objects than the experiments. The controlled results still assume finite active contexts and alphabets, a bounded horizon, known licensed kernels, and an observed state sufficient for legality, costs, and future branch laws. They do not solve estimated-kernel coverage, infinite-horizon free cycles, or the existence of a separating legal policy. The remaining general results do not establish how often mixed fibres occur in natural scientific work or whether a language model can identify the required order and hard predicates. Conversely, the exact substitution result rules out any claim of inherent expressivity once a donor interface is sufficient for the same decision. The empirical question is therefore whether real donor interfaces collapse contexts requiring different scientific transitions, not whether the envelope can defeat a semantic copy of itself.

Claim authority is hierarchical rather than cumulative.

Level	Current evidence	Authorized conclusion
Analytic	Envelope, factorization, deficiency, conservative-front, controlled-state, transcript, and rectangularity results	Conditional mathematical statements under their assumptions; donor-derived tools do not establish empirical prevalence or generic novelty.
Mechanical	Two frozen 2,882-world runs with protected host-side matrices and direct ablations	The typed licensing conjunction works on the registered mechanical generators; host protection is not external custody.
Authored exact contract	A comparator-fair 400-case witness plus 200 interface-order substitutions	The declared licensing relation and its information ablation are separated on those exact contracts; an information-equivalent product must tie.
Development/ diagnostic	The unsupported historical test, comparator-defect chronology, leak audit, and metadata preflights	Design defects, shortcut risks, and execution blockers are identified; no performance transport follows.
Protected transport	An outcome-blind R7A protocol with 28 metadata roots, but zero admissible clusters, zero ready arms, and no custody bindings	Design and fail-closed preflight only; no candidate or comparator outcome is authorized.
External authority	No owner-separated source, gold, execution, scoring, verification, or peer-review receipt	Naturalistic benefit, model generality, independent replication, and top-tier acceptance remain undetermined.

Recent autonomous-research benchmarks report substantial remaining difficulty in wide literature discovery and end-to-end scientific rediscovery [43, 44]. Those results are treated here as motivation for hostile external evaluation, not as evidence that this problem is solved.

12.2 Boundaries of the evidence

Formal generality is not empirical generality. The envelope theory is not tied to $K/W/M$, five gates, or finite generated worlds. The reported effect sizes are. Population claims about biomedical, environmental, scientific-software, or physical-engineering practice require the source-clustered naturalistic successor and cannot be obtained by restating a theorem.

The first outcome-blind source-metadata preflight fixed twelve official GitHub software repository identities. A provider-diverse successor now fixes 16 candidate roots across two waves; each wave has four metadata providers, four artifact modalities, four nominal domains and two roots per domain, with no cross-wave family or persistent-identifier collision. It opened no issue, article, dataset or case text and assigned no pair roles or protected decisions. The successor passes 13/13 metadata-conformance checks and 16/16 live identity checks, but exact historical Crossref bytes are unavailable for eight roots, content-class rights are unresolved for all sixteen and case eligibility is unassessed for all sixteen. It is therefore a structurally diverse acquisition candidate frame, not the required 384 paired plus 64 unresolved clusters in each of two source-disjoint waves. Its scientific terminal remains `P1_R7A_READY_FOR_EXTERNAL_BINDING_NOT_A_SCIENTIFIC_RESULT`.

A separate pinned Retraction Watch–Europe PMC successor closes one narrower source-universe question. Its seven-field outcome-blind gateway admits 49,878 explicit CC0 metadata relations in 42,924 connected families, and exact allowlisted licences exist on both endpoints for 12,038 relations in 11,602 families. Three provider-native metadata strata satisfy the frozen two-wave width gate. The census assigns zero scientific terminals and opens no case text, action/outcome column, model output, or protected score. Publication type is not responsibility gold; the earlier construct-validity terminal therefore remains active.

A further structured-action successor makes that boundary empirical rather than merely cautionary. It assigns 11,976 of 11,991 rights-valid notice records to exactly one of the three provider-native strata and obtains normalized relation-type agreement on 11,988 notices, but satisfies zero of the twelve frozen owner-algebra requirement groups. The later public-standard successor strengthens the available scaffold: eight official documents from four institutional families supply structural analogues for nine groups. Yet no group has named-custodian authorship or explicit delegation, so the exact nondelegation upper bound remains zero sufficient groups. A provider relation or standards token is not an exhaustive R7 target coordinate or operation licence, and item-level or document-level rights are not rights to a completed target algebra. Closed-world semantics, forbidden operations, authority, non-success behavior, signed owner ratification and independent semantic review remain absent. Thus neither the large cross-provider frame nor the stronger standards envelope can itself license a scientific transition.

A comparator family count is not an external-arm freeze. The live binding audit found Agent Laboratory to be the cleanest MIT-licensed open candidate and AI Scientist-v2 to be a separate custom-licensed pressure test. Neither currently consumes R7's byte-equivalent dossier and registered action interface, and neither emits its protected responsibility decision under matched provider, model, tool, time and cost limits. The nine-family successor count therefore has no exact external system identity or lawful adapter yet; its current external-slot terminal is `P1_EXTERNAL_SLOT_CANNOT_CHECK_INTERFACE_BINDING`. A local installation or workflow trace cannot substitute for the missing interface-fair comparison.

A set-theoretic map is not yet a semantics-preserving adapter. This source-terminal transport problem is separate from the external-system adapters above. An outcome-blind audit enumerated all $7^6 = 117{,}649$ maps from the frozen six-terminal postpublication source taxonomy to the seven R7 decision identifiers. None is presently fully certified, but 720 injective maps remain not disproved and semantically unresolved; zero certified maps therefore does not mean zero possible maps. The only currently certified authority-safe partial mapping preserves Unresolved. Every actionable source terminal remains ambiguous because the target decisions lack exhaustive licensed-operation, forbidden-operation and postpublication-coordinate denotations, and no owner-approved bridge equates those decisions with the separately typed diagnostic actions. Captured JATS, MeSH, Crossref, NISO and COPE materials distinguish several correction, retraction, notice and removal relations, but create neither an R7 handle nor authority for a case transition. Full artifact withdrawal has no target image only under a proposed closed-world target semantics whose owner-ratification is currently missing; the resulting impossibility witness is conditional, not an unconditional result about R7.

The subsequent owner/custody sequence changes the execution interface, not this scientific boundary. V11 binds a genuine pre-cutoff NISO standards-governance record, but neither its issuer nor its governance bodies supply R7 ownership, delegation, target rights, a complete seven-target algebra, adapter-blind custody, or independent semantic review. V12 binds the public issue chronology, but the issue contains 0/7 frozen formal-delegation fields and the two exact-commit public routes for the root licence and V8 schema each return HTTP 404; consequently 0/4 authority acts close. V13 replaces further metadata search with an executable, content-addressed unsigned packet: eight administrative components and the exact three-file V10 semantic allowlist are byte-bound, and RFC 8785 signing domains remove the earlier self-hash/review cycle. No external identity, signature, right, custody or review is thereby created. With 0/7 signed external outputs received and 0/4 authority acts closed, the audit remains 116,929 rejected, 720 Cannot-check, and zero certified maps. Packet integrity and signing readiness are therefore provider execution readiness only, not R7 authority or a scientific positive.

Exact contract families are not external science. The mutation-necessity experiment uses fresh pre-bound seeds, protected matrices, three strong assimilated parents, and separate-code-path rescoring, but its worlds are generated mechanical response environments. The revision-responsibility battery broadens the *types of revision contract* tested across five exact families, not the perceptual or linguistic realism of the agent. Its comparator-fair execution and interface-order substitutions show that the tested escalation contract can matter once donor mechanisms are granted, but they do not show that a language-model research agent will correctly infer the same responsibility classes from ambiguous natural scientific evidence. Independently learned diagnosis and recovery engines, noisy observations, open-web evidence, and deployed scientific workflows remain untested.

The meta-discovery results are narrowing results. The bounded study programme freezes a method for asking whether a proposed epistemic mechanic has incremental value; it does not establish a universal basis of atomic mechanics. Its first three broad labels each decompose into several candidates. The scoped failure-memory study finds a useful representation of negative knowledge but no incremental mechanism beyond strong standard parents. The protected zero-error representation-transition study is stronger evidence and still negative for this architecture's ownership: the full candidate ties the verified regime-revision parent on both disjoint frozen splits. None of these results licenses an automatic $K/W/M$ mutation or a general scientific-creativity claim.

Ideal-product equivalence is a substantive boundary. The escalation contract has no inherent expressivity advantage over a donor product supplied the same responsibility classes, partial order, counterfactual results, preservation and reopen predicates, and authority semantics: that product ties exactly. Any practical advantage therefore depends on the value of making this scientific escalation interface explicit in realistic systems, not on a claim that centralization creates additional semantic power.

Negative execution history matters. The first execution of the revision-responsibility battery is retained because an independently discovered post-access defect in two comparators weakened its clean-control comparison. The defect was not patched after outcomes had been seen; the publication-authorizing comparison is the separately frozen, disjoint second execution. This protects the chronology, but it also emphasizes that the wider claim rests on exact contract engineering that still needs external validation.

Reconstruction quality is only partially measured. The local suite checks key transition behaviours, but the deeper hypotheses about late-depth trace fidelity, reviewer localization time, and semantic mismatch require a dedicated prospective protocol.

Adjacent capabilities are delegated, not solved. Route coverage and recall for open-world search, scientific semantic projection across sources, evidence-binding and protected verification, and the evolution of the method state itself are all separate problems with their own evidence requirements. A success here is not evidence about any of them. In particular, the mechanisms studied here can expose a possible method failure but do not authorize M_t to rewrite itself.

Literature closure is bounded rather than universal. The nearest-work rounds reached bounded fixed points for their exact residuals and directly shaped the strongest parents. Because the repair-agent literature is moving rapidly, a final pre-submission delta search and replacement of preprints by archival versions remain required. No broad novelty statement follows from a bounded fixed point.

12.3 Observed limitations

The hidden-formulation study remains underpowered. With $N = 48$ cases its achieved half-width is 0.1414, short of the lowest registered precision tier, and it is underpowered for its original broad primary hypothesis. Those records, including the low root-success rates in Tables 2 and 3, are retained as negative history and are not pooled with the powered successor. The mutation-necessity experiment addresses a narrower prospectively frozen claim on 2,882 worlds per run; it does not retroactively validate the earlier broader formulation.

Perfect success is generator-bounded. The governed policy attains protected success 1.0000 in both mutation-necessity runs, and the escalation contract attains 400 of 400, an exact rate of 1.000, on its exact contract battery. These ceilings are credible only at the level of their deterministic registered generators and explicit action and response vocabularies. Strong parents, hostile ablations, the comparator-fair donor product, and the ideal-product tie make the tested distinctions non-vacuous, but the ceiling results may not survive perceptual ambiguity, noisy interventions, learned proposal generation, or imperfect responsibility inference.

Template-level leaks exist in the local suite. The solvability audit identified perfect separation on fixed phrases—"The team's framing is" distinguishes all 44 hidden-shift cases from all 22 controls—and filename-based family separation, in which proposal and trial stems identify decomposition cases. These leaks were documented but not repaired in this iteration.

Mechanical arm generality is untested. Half the correct attributions, 32 of 66, rest on rules that fire on exactly one case. Only four general relations cover three or more cases. A fresh set of naturalistic cases is required to distinguish general mechanisms from case-specific recognizers.

Blind-responder shortcuts affect half the suite. Thirty-three of 66 cases can be solved by a classifier that reads only resource path stems and two fixed template phrases. The remaining 33 parent-domain, representation and measurement cases require genuine arithmetic reasoning.

13 Data availability

Every artifact named anywhere in this paper is listed here with its path, so that no repository location has to be carried in the narrative.

13.1 Preregistration

The study was preregistered and frozen before any outcome was accessed, under the identifier `P1.hidden-formulation.v1`. A reader checking whether a reported analysis was planned or chosen after the fact should compare against that freeze; it is the reason every number reported here can be traced to a decision made in advance of seeing it.

The machine-readable freeze is `protocol/PROTOCOL_V1.json` and the case requirements are `protocol/HIDDEN_SHIFT_CASE_SCHEMA_V1.json`. The freeze records that no outcome had been accessed when it was cut. The amendment that replaced human adjudication with a blinded model panel is recorded as version 1.1 of that freeze; it carries an adjudication tier that the record type refuses to report as human adjudication. The adjudication standard itself is `protocol/ADJUDICATION_RUBRIC_V1.md`, the execution identities are in `protocol/EXECUTION_MANIFEST_V1.md`, and the prospective power calculation is `protocol/PROSPECTIVE_POWER_V1.md`.

The frozen experiments reported in Sections 6 and 7 have their own preregistrations: `research/revival/p1/protocol/P1.epistemic-mutation-necessity.v2.2.4.json` for the mutation-necessity experiment, together with its superseded earlier versions in the same directory, and `research/claim_expansion/p1/P1_X_PROTOCOL_V1.md` with `research/claim_expansion/p1/v2/P1_X_V2_REPLICATION_PROTOCOL.md` for the revision-responsibility battery.

13.2 Frozen suite and generated quantities

The case suite is `protocol/cases/`, split into a pilot and a test set and bound by content hash: the pilot fingerprint is `7a50a2d5025b` and the test fingerprint is `21b461d89280`. Case generation and validation are `protocol/cases/generate_cases.py` and `protocol/cases/validate_cases.py`. Every count, tier and margin quoted in the text is a macro generated into `manuscript/generated/suite_facts.tex` by `scripts/render_suite_facts.py`, so prose cannot drift from the frozen suite. The precision tier the frozen power record assigns to the achieved half-width is registered there under the label `BELOW_TIER_D`, with 385 cases required for the primary hypothesis and 2,401 for the restraint hypothesis. The tracked `results/INTEGRITY_NOTE.md` preserves the failed historical attempt at commit `85d32db0`: 2,794 scored records and 86 CANNOT-CHECK records, all in `orion_live_provider` (78 authentication, four rate-limit and four internal-server errors), with 23 of 48 provider cases lacking a scored repeat. That 86-record count describes a historical execution failure, not current missingness, and no per-cell rerun manifest is available in a clean checkout. The current tracked raw archive is authoritative and contains 2,880 of 2,880 `OK` run records and 2,880 of 2,880 `SCORED` score records, with zero current CANNOT-CHECK records.

13.3 The hidden-formulation study

The scored archive is `results/raw/test_scored.jsonl` with its run records in `results/raw/test_runs.jsonl`. Table 2 projects `results/P1-T2_baseline_ablation_results.json` and its Markdown companion, which also carry the full system identifiers, per-family results and mechanistic metrics; Table 3 projects `results/P1-T3_failure_taxonomy.json` and its Markdown companion, which carry the per-case detail behind the failure-mode roll-up. The integrity note for these records is `results/INTEGRITY_NOTE.md`.

The mechanical attribution result and its rule-level limitations are `evidence/MECHANICAL_ARM_LIMITATIONS_V1.md` with the case table `evidence/MECHANICAL_ARM_CASE_TABLE_V1.json`. The blind solvability audit is `evidence/MECHANICAL_SOLVABILITY_AUDIT_V1.md` with `evidence/mechanical_solvability_audit_v1.json`. The falsifier record, including the subject commit `8a8a7feed588363f8e2cd820d3399a33b7af3074` and continuous-integration run `31933432314` at which the suite passed, is `evidence/FALSIFIER_V1.md`. The bounded parent-domain replay is `evidence/PARENT_DOMAIN_REPLAY_V1.md`, and the saturation record for the suite is `evidence/FINAL_SATURATION_AUDIT.md`.

13.4 The mutation-necessity experiment

The primary and replication directories under `research/revival/p1/confirmatory/v2.2/` each contain compressed public worlds, protected response matrices, raw results, the result record, an separate-implementation verification record, provenance, and a `SHA256SUMS` manifest. The primary execution identity is `PRIMARY_EXECUTION_FREEZE_V3.json`; earlier receipts and both additive amendments are retained beside it. The replication directory binds its own disjoint world and execution receipts. `PRIMARY_REPLICATION_CONCORDANCE.json` checks the predeclared cross-run conditions without pooling the runs. The separate verifier is `research/revival/p1/verify_mutation_necessity_independent.py`; it uses only the standard library and imports neither the candidate policies, the production scorer and statistics, nor the subject's licensing code. This supplies code-path separation within the repository, not external scorer custody.

The round-by-round donor dispositions that fixed the parent and ablation structure are `research/revival/p1/P1_DONOR_ASSIMILATION_LEDGER_V1.json` with its later rounds in the same directory, `research/revival/p1/P1_DONOR_ENGULFMENT_V1.json`, and the saturation record `research/revival/p1/literature/SATURATION_LEDGER.md`.

13.5 The revision-responsibility battery

The protected freeze, protected results, and separate-implementation verification for both executions are under `research/claim_expansion/p1/`: the first execution in `results/P1_X_PROTECTED_RESULT_V1.json` with its outcome-access receipt, and the comparator-fair disjoint second execution in `v2/P1_X_PROTECTED_RESULT_V2.json` with `v2/P1_X_INDEPENDENT_VERIFICATION_V2.json`. The case schema is `research/claim_expansion/p1/SCIENTIFIC_REVISION_RESPONSIBILITY_CASE_V1.schema.json`.

13.6 Bounded meta-discovery studies

The study calculus and the three decomposition pilots are under `development/recursive-atom-calculus-closure-v1/` and `development/jump-atom-pilot-closure-v1/`. The scoped failure-knowledge study is `development/failure-epistemology-prospective-closure-v1/`, and the protected zero-error representation-transition study is `research/extensions/orion-jump-recursive-atoms/zero_error_jump/`.

13.7 Claim ledgers and nearest work

The mapping from each result-bearing statement in this manuscript to the archived artifact that supports it is `evidence/CLAIM_LEDGER.md`, with the immutable record of the earlier revision at `evidence/CLAIM_LEDGER_V1.md`. The full nearest-work mechanism matrix, from which Section 10 is drawn, is `evidence/NEAREST_WORK_MATRIX_V2.md` with `evidence/nearest_work_matrix_v2.json`. The bounded readiness attestation is `evidence/PEER_REVIEW_READY_BOUNDED_V2.md`.

13.8 Naturalistic and comparator preflights

The outcome-blind metadata snapshot used to prepare the R7A successor is `development/naturalistic-panel-harness-2026-08-23/SOURCE_METADATA_SNAPSHOT_V1.json`, with canonical payload digest `7dd283acd81e63e84887d82ee7761e89e62a47d31afceca8628e456b76f73e01`. Its protocol-binding status and exact non-promotion boundary are recorded in `development/naturalistic-panel-harness-2026-08-23/PROTOCOL_BINDING_STATUS_V1.json`. These files contain repository/revision/licence metadata only and do not contain R7 cases, labels, pair assignments, scorer outputs or outcomes. The provider- and modality-diverse successor frame is `development/provider-diverse-metadata-census-2026-08-23/SOURCE_CENSUS_V1.json`, with its frozen rules in `development/provider-diverse-metadata-census-2026-08-23/CENSUS_PROTOCOL_V1.json` and its live conformance receipt in `development/provider-diverse-metadata-census-2026-08-23/VERIFICATION_RECEIPT_V1.json`. It likewise contains no case or outcome authority; historical-byte, content-rights and eligibility terminals remain explicit.

A prospectively frozen public correction/retraction source-feasibility probe is under `development/p1-naturalistic-transport-successor-2026-08-23/`. The protocol, count-only Europe PMC receipt, Crossref rights amendment, source audit, and checksum manifest are preserved there. The pinned Retraction Watch CSV itself is not redistributed in this repository; its receipt records 71,944 rows and the byte digest `ceaab201d728dfcf9929ec1e229acd2ad88c650c847ec922ba9ffe831e366abb`. No Europe PMC result row, relation, case text, pair assignment, typed-action gold item, system output, or protected score was opened or created by that probe. The official CC0 binding applies to the database metadata only, not to linked article, notice, attachment, abstract, supplement, or blog content.

The outcome-blind rights/relation successor is preserved under `development/p1-rw-rights-relation-census-v2-2026-08-23/`. Its exact Retraction Watch relation census, bounded Europe PMC rights reducer, source-native aggregate strata, amendments, negative ledger, validation receipt, and 24-entry checksum manifest are retained there. The raw 65,984,968-byte transport is not redistributed; its SHA-256 is `ceaab201d728dfcf9929ec1e229acd2ad88c650c847ec922ba9ffe831e366abb`. The deterministic reducer reports 49,878 admitted relations in 42,924 families, including 12,038 exact-both-endpoint-rights relations in 11,602 families. It accessed no case text, scientific terminal, model output, or protected score. Missing provider matches remain CANNOT-CHECK; they are not incompatible-rights or source-absence findings.

The structured-action semantics successor is `development/p1-structured-action-semantics-v3-2026-08-23/`. Its frozen protocol and parser, immutable raw-interface failure, prospectively frozen normalization amendment, normalized provider census, twelve-group owner-field feasibility record, rights/provenance receipt, construct-validity failure atlas, negative ledger, result, validation receipt, and checksum manifest are retained there. The result digest is `7178052320f85ea112e4b0b78da558fc5a68f7dc3b465a4f7afc9e1915860f2b`. The packet retains aggregate token frequencies only: no identifiers, raw provider responses, case text, scientific gold, model output, or protected score. Its 52 scientific-artifact assertions pass. The initial one-of-11,991 raw interface match is retained as a serialization failure rather than overwritten; only the separately frozen display-token normalization supports the 11,988-of-11,991 relation-set agreement result.

13.9 Public-standard owner-algebra feasibility

The outcome-blind standards packet is `development/p1-owner-algebra-construct-validity-v4-2026-08-23/`. Its protocol and parser-only amendment were frozen before the corresponding captures. The retained artifacts bind eight HTTP-200 documents from four institutional families and one fail-closed HTTP-404 route; raw source documents were deleted. The packet-native validator checks 85 scientific assertions without network access, repository CI, case text, outcomes, or protected scores. Its result records 9/12 structural analogues, 0/12 named-custodian or delegation groups, 0/12 sufficient groups, and zero scientific-action gold. The validator and 18-entry manifest establish bounded internal consistency and byte integrity, not independent semantic review or owner authority.

13.10 Outcome-blind source-native adapter audit

The full 47-entry semantic-interface handoff and its recorded independent checksum receipt remain in external custody and are not present in a clean repository checkout. The manuscript retains only the recorded handoff manifest digest, `9f4503f693e155b12fb8c333f4777619cec1d66e387dcfc0e141fffa6933847d`. Because neither the manifest entries nor the independent receipt are repository-available, clean-checkout byte verification is CANNOT-CHECK; the historical zero-failure checksum statement is not a current repository integrity result. No private or large payload is redistributed here, and the unavailable handoff cannot serve as repository-available evidence or external semantic authority.

The audit enumerates all 117,649 total maps: 0 are fully certified, 116,929 are rejected, and 720 remain not disproved but uncertified. The only currently certified partial map preserves UNRESOLVED; the actionable mappings remain unbound because exhaustive target denotations and an owner-approved decision/action bridge are absent. The full-withdrawal counterexample is explicitly conditional on later owner-ratification of proposed closed-world target denotations. No case text, row-level outcome, protected gold or system outcome was accessed. These counts are bounded provenance statements recorded from the unavailable handoff, not independently checkable current-repository facts or external semantic authority. Licensing for the captured public taxonomies remains uncertain; no redistribution right or R7 transition is assumed.

The external comparator identity and interface audit is `development/comparator-binding-harness-2026-08-23/COMPARATOR_BINDING_LEDGER_V1.json`, with its scientific interpretation in `development/comparator-binding-harness-2026-08-23/SCIENTIFIC_REPORT.md`. Its cross-paper terminal is `CANNOT_CHECK_FULL_COMPARATOR_FREEZE`; bounded installation or entrypoint probes in that harness grant no P1 performance or external-authority claim.

13.11 Owner/custody execution packets V11–V13

The historical-governance audit is preserved under `development/p1-owner-custody-positive-successor-v11-2026-08-24/`. Its `SHA256SUMS` manifest digest is `8ebd874f0b0a7f3a3a7877c698dee1371352ebc4ce5dc3bb1a1bd2a2d0629721`. The public issue-chronology, delegation-field and exact-commit route audit is preserved under `development/p1-owner-custody-positive-successor-v12-2026-08-24/`. Its manifest digest is `7a136154d7fbc3c52d51ea2dade7829446842c21381f3c9c912189d47dc29407`. The content-addressed external-execution packet is preserved under `development/p1-owner-custody-positive-successor-v13-2026-08-24/`. Its manifest digest is `c2ff7a87356559bfce4ea4033aae807a6a5c4ce03b8b43c29782b58c29d33667`. These three manifests verify the retained packet bytes only; they do not supply external identity, delegation, signatures, rights, custody, review, map certification or scientific outcomes.

Exact V13 terminal:
`P1_V13_CONTENT_ADDRESSED_EXTERNAL_EXECUTION_PACKET_COMPLETE__ZERO_OF_SEVEN_EXTERNAL_SIGNED_OUTPUTS_RECEIVED__ZERO_OF_FOUR_AUTHORITY_ACTS_CLOSED__720_MAPS_UNCHANGED`

14 Code availability

14.1 Current review source and historical package identity

The current review source is not yet bound to a public archive URL, archive DOI, immutable submission commit, or verified repository-level licence. Editors and reviewers must therefore receive the exact current snapshot before external review. The PDF under `journal_package/` remains a content-bound record of an earlier inspected manuscript, not a rendering of this revision: `journal_package/RENDER_CLOSURE_STATE.json` marks that package `SUPERSEDED`. It must not be relabelled or submitted as the current PDF; the current source requires a fresh immutable release, build, and visual audit.

14.2 Repository access and reproduction

The implementation repository holds the source code, protocol definitions, case suite, and repository-held evidence artifacts used by this revision. Persistent DOI assignment remains pending publication. Result-bearing artifacts are identified by their repository paths and content bindings rather than by an unverified archival claim.

The exact decompression, independent-rescoring, byte-comparison, and figure-regeneration commands are given in `REPRODUCE.md`. The command

```
make paper01-results
```

regenerates the publication tables from archived raw records only. It does not execute systems under test or create missing external evidence. On a checkout without an admissible archived study run the underlying module reports the result as undetermined; malformed or unbound archives are refused rather than converted into publishable numbers.

A fresh code and test environment requires Python 3.11 or later, an editable install with the development extras from the repository root, and model-provider access only for commands that actually execute a live-provider arm; dependencies are declared in `pyproject.toml`.

14.3 Runtime and cost

The credential-free mutation-necessity campaign and its independent verification each complete in approximately one minute on the current CPU environment. They make no network call and incur no model-provider cost. Algorithmic effort is fixed at 2,882 worlds, nine runnable arms, five ablations, and 10,000 registered paired-bootstrap resamples. The provider cost of the earlier live-provider arm was not recorded under a bound environment and therefore remains undetermined; it is not imputed from the mechanical successor.

14.4 Licensing status

This revision does not contain a repository-level licence file or a project licence declaration in `pyproject.toml`. Redistribution terms for code, case definitions, and protocol artifacts therefore remain undetermined from the repository state and must be resolved explicitly before archival or publication. Model-provider terms remain separately governed by the relevant providers.

14.5 Artifact fingerprinting

Result-bearing execution identities are bound through the protocol and archived records. Relevant coordinates include dataset and suite revisions and fingerprints; baseline configuration hashes; evaluator identity; frozen split hashes; and the exact subject revision and provider/model revisions for a live run. Reproduction of an empirical result requires the exact bound subject and admissible archived records; prose or a protocol declaration alone is not evidence that the run occurred.

15 Ethics, safety, and resources

15.1 Ethical considerations

This work studies mechanistic aspects of research workflow design rather than reporting a deployment in a human-facing domain. The frozen local evaluation uses constructed cases and does not require human-subject records or personally identifiable information. Any later external-provider or retrieval execution must be documented separately from this local evidence rather than inferred from the protocol design.

15.2 Safety considerations

Authority promotion and formulation reconstruction are separate concerns, and only the second is studied here. The mechanisms described can expose a possible method failure but cannot authorize M_t to rewrite itself. The reframe gate is mechanically bounded: only formulation and search-universe coordinates may trigger a reframe, and the evidence-only and execution-only negative controls exercise the corresponding non-reframe behaviour in the frozen local suite.

15.3 Resource usage and environmental impact

The repository does not yet contain an admissible clean-environment resource ledger for the full campaign. Numerical model-token, runtime, monetary-cost and carbon-impact quantities therefore remain undetermined in this revision. A future result-bearing run must retain actual resource usage under its frozen provider and environment bindings before those quantities are reported.

15.4 Dual-use considerations

The epistemic-state machinery described here is general-purpose and could be applied outside scientific research. This paper evaluates scientific-workflow mechanics and does not present an empirical evaluation of military or other high-risk deployment applications. The authority boundaries above are part of the implemented design, but they should not be read as evidence of safety in domains that were not evaluated.

16 Conclusion

This paper asks which scientific layer evidence authorizes changing after diagnosis and repair mechanisms have already produced candidates and counterfactual outcomes. Its science-specific proposal is typed responsibility-to-authority licensing: causal responsibility, successful intervention, and justified revision are kept distinct, and a broader transition is admitted only after narrower alternatives, same-task effect, protected invariants, dependency impact, and budget have been checked.

The epistemic transition envelope is the formal vehicle for this proposal, not a claim to have invented general information sufficiency. Exact donor factorization states when another interface implements the same licensed decision and therefore must tie. Legal-action deficiency quantifies declared loss only when a fibrewise legal policy exists. Conservative-front and universal-representation results delimit the formalism. The controlled-state frontier, transcript-support condition, and rectangularity boundary are explicitly donor-derived audit tools rather than new POMDP, active-testing, or multiple-priors mathematics.

The resulting hierarchy is essential. The analytic statements range over broad mathematical objects. The empirical results establish only registered mechanical and authored exact-contract behaviour. The development falsifiers identify defects and shortcuts. The protected naturalistic study is not yet executable, and no external evaluation or peer-review authority has been obtained.

The powered mutation-necessity experiment supports that bounded internal composition on its credential-free mechanical family. Across a prospectively frozen primary run and a disjoint replication, the governed policy attains protected hidden-shift success 1.0000, preserves perfect control restraint and protected success, and passes every frozen non-regression, safety, multiplicity and cost gate. The comparative reading of that result is withdrawn: a faithful comparator granted the same ordered search reaches 1.00000 with zero forbidden mutations and matches the governed policy on both registered components. Direct ablations reproduce the intended mechanism failures, and a separately written implementation recomputes every score and analysis field without disagreement. This is local implementation separation, not external evaluation authority.

The revision-responsibility battery widens the evidence from one hidden-shift mechanism to the control problem of scientific revision responsibility. In a separately frozen, comparator-fair disjoint execution, the escalation contract achieves 400 of 400 exact revision decisions versus 275 of 400 for a donor-complete greedy product, a paired advantage of 0.3125 with 95% CI [0.2675,0.3575], while preserving zero false high-level reframes and zero protected-invariant violations. The effect is positive in every exact domain and survives the tested equally informative diagnosis and recovery ordering substitutions. A separately written implementation reproduces the protected identities, result hashes and headline counts.

The historical predecessor remains an adverse result rather than supporting context: its broad primary test was not supported, the subject and registered strong baselines each achieved only 1 of 48 root successes, 32 of 66 attributions used singleton rules, and phrase/filename cues enabled a blind responder on 33 of 66 cases. These observations motivate the naturalistic responsibility-inference successor; they are not overwritten by the later exact mechanical results.

The strongest empirical boundary is equally important: an ideal donor product supplied exactly the same scientific responsibility, minimality, preservation, reopen and authority semantics ties at 400 of 400. The factorization theorem requires this tie. The evidence therefore isolates one deliberately lossy donor interface; it does not establish an inherent centralization advantage. The next scientific target is to estimate legal-action deficiency and decision-mixed fibre prevalence under source-clustered naturalistic acquisition, then determine whether each interface-fair external system already exposes a sufficient statistic. Twenty-eight exact metadata roots are pinned across the provider-diverse and repository seed frames. Separately, a pinned public-source census establishes 12,038 exact-rights relations in 11,602 connected families and three two-wave metadata strata. A structured-action successor assigns 11,976 of 11,991 unique notices to exactly one stratum and obtains normalized relation-type agreement for 11,988 notices, but zero of twelve frozen owner-algebra requirement groups is sufficient. A separate byte-bound four-institution standards envelope supplies structural analogues for nine groups, but named-custodian authorship or delegation for none. This is a rights-bounded, provider-conformant development scaffold, not a construct-valid panel. The source-terminal audit remains exact and adverse: 116,929 of 117,649 maps are rejected, 720 remain Cannot-check, and zero are certified. V11 binds historical standards governance without R7 authority; V12 binds issue chronology but 0/7 delegation fields and two exact-commit HTTP-404 routes; V13 binds an unsigned external-execution packet but has received 0/7 signed outputs and closed 0/4 authority acts. The next discriminator is therefore external signed execution and deterministic bundle acceptance rather than another metadata search. That operational advance is not scientific closure. Zero scientific-action gold assignments, zero eligible R7A case dossiers, zero ready primary comparator arms, and zero external custody rows currently authorize execution. Until the two source-disjoint waves, comparator interface and independent scorers are bound and executed, model-general, open-ended, retrieval, and deployed autonomous-science superiority remain undetermined.

The bounded meta-discovery programme sharpens the same donor-first lesson. Its study calculus survives, but the first broad labels for tension, thought experiment and concept each split into several candidate mechanics rather than becoming universal new primitives. Scoped failure knowledge is useful while standard failure-learning parents remain sufficient for the tested incremental effect. Most decisively, on the protected zero-error representation-transition study the candidate composition and the strongest verified regime-revision parent tie exactly on both disjoint frozen splits, including zero false transitions. Those outcomes are recorded as decomposition and parent subsumption: they improve the boundary on when to investigate or reopen a formulation, but they do not add a new general $K/W/M$ mutation operator or a scientific-creativity claim.

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